

Department of Electronics and Communication Engineering

UNIVERSITY OF VISVESVARAYA COLLEGE OF ENGINEERING, KR CIRCLE, BENGALURU,
KARNATAKA – 560001

Research Internship Report

KRITHIK KUMAR

(U25UV24T043048)

Under the guidance of

*Prof. Prabu K, Associate Professor, Department of Electronics and Communication
Engineering, National Institute of Technology Karnataka (NITK), Surathkal*

Internship Organisation: National Institute of Technology Karnataka
(NITK), Surathkal (Educational Institution)

Duration of Internship: 21 May 2026 – 19 July 2026 (Total: 60 Days)



SUBMITTED BY

KRITHIK KUMAR

4TH SEMESTER, ELECTRONICS AND COMMUNICATION ENGINEERING.
UNIVERSITY OF VISVESVARAYA COLLEGE OF ENGINEERING (UVCE), BENGALURU

ACKNOWLEDGEMENT

I wish to express my sincere gratitude to Prof. Prabu K, Associate Professor, Department of Electronics and Communication, National Institute of Technology Karnataka, Surathkal, for accepting me as a research intern and for the guidance offered throughout the internship. The regular discussions, the reading list that was suggested, and the critical feedback on my interpretation of research papers shaped both the direction and the rigour of this work.

I am grateful to the research scholars and members of the laboratory at NITK for the informal but valuable technical discussions on cooperative perception, wireless communication, and simulation methodology, and for their patience in answering questions that must often have seemed elementary.

I thank Dr Kiran K, Head of the Department of Electronics and Communication, University of Visvesvaraya College of Engineering and Prof. Subhasish Tripathy, Director, University of Visvesvaraya College of Engineering, for permitting me to undertake this internship and for the institutional support extended to me.

ABSTRACT

Vehicle-to-Everything (V2X) communication enables a vehicle to exchange information with other vehicles, with roadside infrastructure, with vulnerable road users and with network services. It is regarded as a key enabler of Intelligent Transportation Systems and of higher levels of driving automation, because it allows a vehicle to act on information that its own sensors could never have obtained.

This report documents a study-oriented research internship undertaken in the Department of Electronics and Communication, National Institute of Technology Karnataka, Surathkal, over a period of 60 days. The work consisted of five strands. The first was a study of V2X fundamentals: the four communication modes (V2V, V2I, V2P, V2N), the standardised message sets, and the two radio access families DSRC based on IEEE 802.11p, and Cellular V2X based on 3GPP LTE-V2X and NR-V2X together with their evolution towards 6G. The second was a study of the impairments that characterise vehicular links, including delay, jitter, packet loss, clock drift, interruption and bandwidth limitation, and of their physical causes. The third was a study of cooperative perception: why vehicles collaborate, at what level they exchange information, how the perception pipeline is organised, and what role machine-learning methods play within it. The fourth was familiarisation with the simulation frameworks (SUMO, OMNeT++, Veins, ns-3, CARLA, OpenCOOD) and public datasets (OPV2V, V2XSet, DAIR-V2X) on which the field depends, together with a small comparative simulation exercise contrasting the medium-access behaviour of DSRC and C-V2X. The fifth was a review of V2X cybersecurity, arranged as a taxonomy running from spoofing to denial-of-service, and of the countermeasures standardised or proposed against it.

The internship did not produce a new algorithm; its outcome is a consolidated understanding of the problem space, a working familiarity with the tools and datasets used in it, and a clear view of where the open problems lie.

TABLE OF CONTENTS

	Title	Page
1	Introduction	1
2	Fundamentals of V2X Communication	5
3	Communication Challenges in V2X	10
4	Temporal Challenges in Cooperative Perception	14
5	Collaboration in Cooperative Perception	28
6	Machine Learning for Feature Prediction	22
7	Cooperative Perception Architecture	24
8	Evolution from 5G to 6G	27
9	Simulation Frameworks Studied	29
10	Public Datasets	32
11	A Comparative Exercise: DSRC and C-V2X	34
12	V2X Cybersecurity: Threats and Countermeasures	39
13	Internship Outcomes	43
14	Future Work	45
15	Conclusion	46
	References	47
	Appendix A: Weekly Internship Log	A-1

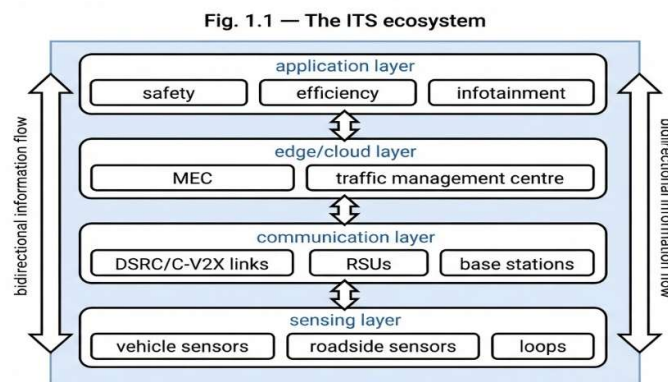
ABBREVIATIONS

3GPP - 3rd Generation Partnership Project	BEV - Bird's-Eye View	BSM - Basic Safety Message	CAM - Cooperative Awareness Message
CAV - Connected and Automated Vehicle	CBR - Channel Busy Ratio	CNN - Convolutional Neural Network	CPM - Collective Perception Message
CSMA/CA - Carrier Sense Multiple Access with Collision Avoidance	C-V2X - Cellular V2X	DENM - Decentralised Environmental Notification Message	DSRC - Dedicated Short-Range Communications
ETSI - European Telecommunications Standards Institute	GNSS - Global Navigation Satellite System	MEC - Multi-access Edge Computing	ITS - Intelligent Transportation System
OBU / RSU - On-Board Unit / Roadside Unit	MAC - Medium Access Control	PKI - Public Key Infrastructure	NR - New Radio
SPS - Semi-Persistent Scheduling	PDR - Packet Delivery Ratio	V2V / V2I / V2P / V2N - Vehicle-to-Vehicle / -	PTP - Precision Time Protocol

1. INTRODUCTION

1.1 Intelligent Transportation Systems

Road transport is one of the few large-scale engineering systems in which most control decisions are still taken by human beings in real time, with no shared information channel between them. The consequences are congestion that grows non-linearly with demand, wasted fuel, and a fatality count that remains stubbornly high. An Intelligent Transportation System (ITS) is the attempt to instrument this system to add sensing, communication and computation to vehicles, roads and traffic-management centres so that decisions can be taken on the basis of information rather than inference.



An ITS is best described as an ecosystem of interacting layers rather than as a single technology: a sensing layer of vehicle and roadside sensors; a communication layer that moves observations from where they are made to where they are needed; an edge and cloud layer that aggregates them; and an application layer signal control, incident detection, cooperative driving that consumes them.

Three properties distinguish an ITS from an ordinary sensor network, and almost every design decision described in this report is a response to one of them. It is safety-critical: a late or wrong message can contribute to a collision. It is highly dynamic: the network topology changes on a timescale of seconds. And it is open: participants are made by different manufacturers, are subject to different regulations, and cannot be assumed to trust one another.

1.2 Connected Vehicles

A connected vehicle is one that exchanges data with entities outside itself. In its weakest form this means a telematics unit reporting diagnostics; in its strongest, a vehicle that broadcasts its own state several times a second, receives the state of every vehicle nearby, and fuses those reports with its own sensor data.

It is useful to separate connectivity from automation. The SAE levels of driving automation describe how much of the driving task the vehicle performs; connectivity describes how much the vehicle knows. The two are orthogonal but complementary: automation without connectivity is bounded by the line of sight of the vehicle's own sensors, while connectivity without automation still helps, because a human driver can be warned of a hazard that is not yet visible.

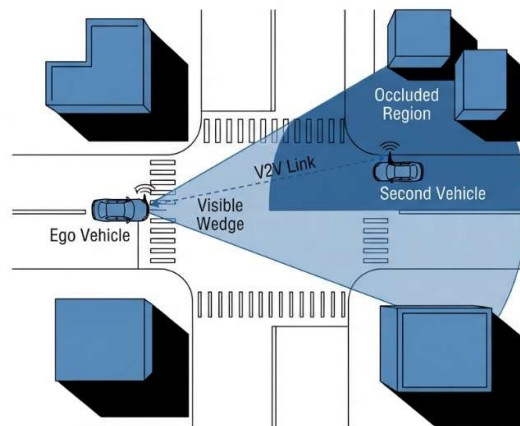
The connected vehicle therefore has a dual role it is both a consumer of information generated elsewhere and a producer of information for the benefit of others. This is the essence of cooperative driving, and it means the benefit each vehicle receives depends on how many others participate and on how faithfully what they transmit reflects reality. The broader vision in which every vehicle is a networked node is often described as the Internet of Vehicles [19].

1.3 The Need for V2X

The case for V2X rests on a simple observation: on-board sensors are line-of-sight devices, and much of what matters in traffic is not in the line of sight.

Consider a vehicle approaching an urban intersection with buildings at the corners. Its LiDAR and cameras see the cross-street only within the wedge of visibility the buildings permit. A vehicle approaching on the cross-street at 50 km/h is invisible until it emerges, by which time the available reaction time may be less than the sum of perception latency, decision latency and braking distance. No improvement in sensor resolution solves this, because the obstruction is geometric, not perceptual. What is required is information from a vantage point that does have line of sight the approaching vehicle itself, or a camera on the traffic signal.

Fig. 1.2 – Perception limits of a single vehicle at an occluded intersection



The argument generalises. Occlusion by heavy vehicles hides pedestrians and cyclists. Fog, heavy rain and low sun degrade sensors for all vehicles simultaneously but the vehicle closest to a hazard usually still sees it. Sensor range is finite, while the distance over which a vehicle at highway speed must anticipate is not small. In each case the missing information exists somewhere in the scene; V2X is the channel that moves it to where it is needed. Beyond

safety, V2X supports traffic efficiency (signal phase and timing broadcast to approaching vehicles), platooning, and cooperative manoeuvring negotiated between vehicles rather than inferred from behaviour.

1.4 Challenges in Autonomous Driving

Autonomous driving is conventionally decomposed into perception, prediction, planning and control. Several open challenges bear directly on the material of this report.

Perception under occlusion and range limits is the binding constraint in dense urban environments, and it is the constraint cooperative perception relaxes. The long tail the rare object, manoeuvre or weather condition on which automation actually fails is poorly represented in training data, and an event that is rare from the perspective of one vehicle is less rare from the perspective of a fleet. Latency budgets are hard: a warning that arrives after the last moment at which the vehicle could have acted on it has no value. Consistency across agents matters because two vehicles acting on different beliefs about the same scene may make manoeuvres that are individually reasonable and jointly disastrous. And trust is a prerequisite: a vehicle that acts on information from another vehicle has extended its trust boundary to a device it does not control, which is a security problem before it is a perception problem (Section 12).

1.5 Motivation and Scope

The motivation for the internship was to obtain a technically grounded understanding of a field that sits at the intersection of wireless communication, computer vision and vehicle safety, and in which the interesting problems arise precisely from that intersection. A recurring observation during the reading and one that shapes the emphasis of Sections 3, 4 and 7 is that much of the cooperative-perception literature assumes an idealised communication link, while the communications literature describes a link that is nothing of the sort.

Within scope. A literature study of V2X communication standards and of cooperative perception; a review of ten recent papers on cooperative perception; a conceptual study of the perception pipeline and of the machine-learning methods used within it; familiarisation with the standard simulation frameworks and public datasets; a review of V2X cybersecurity threats and countermeasures; and a small comparative simulation exercise contrasting DSRC and C-V2X medium access.

Outside scope. No perception algorithm, fusion method or compensation scheme was designed, and no deep-learning model was trained. The simulation frameworks of Section 9 were studied but not used; the datasets of Section 10 were studied through their publications and documentation. These boundaries are restated where relevant.

1.6 Objectives

- O1. Understand V2X communication: the four modes, the message sets, and the radio access technologies that carry them.
- O2. Characterise the communication impairments affecting V2X links and identify their physical causes.
- O3. Understand how those impairments propagate into the cooperative perception pipeline.
- O4. Survey the machine-learning methods used for temporal prediction and feature alignment.
- O5. Review recent research literature on cooperative perception and compare the methods by their assumptions.
- O6. Become familiar with the simulation frameworks and public datasets used in the field.
- O7. Compare the medium-access behaviour of DSRC and C-V2X through a small educational simulation exercise.
- O8. Review the V2X cybersecurity threat landscape and its countermeasures.

1.7 Organisation of the Report

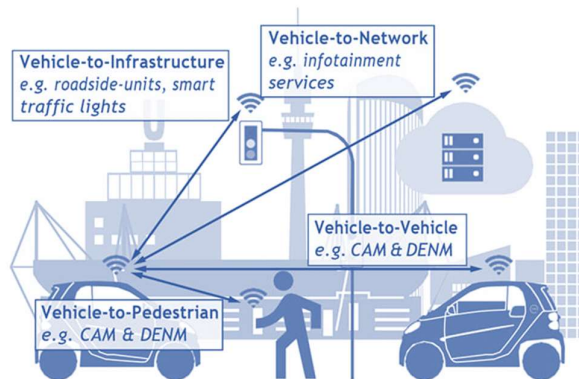
Section 2 introduces the fundamentals of V2X. Section 3 analyses the communication impairments that afflict V2X links. Section 4 carries those impairments into the perception domain. Section 5 asks why vehicles collaborate and at what level. Section 6 surveys the machine-learning methods used for temporal prediction, and Section 7 assembles the cooperative perception architecture. Section 8 looks forward to 6G. Sections 9 and 10 describe the simulation frameworks and datasets studied, and Section 11 reports the small DSRC/C-V2X exercise. Section 12 summarises the cybersecurity review. Sections 13-15 present the outcomes, future directions and conclusions.

2. FUNDAMENTALS OF V2X COMMUNICATION

2.1 Introduction to V2X

V2X is an umbrella term for the communication modes by which a vehicle exchanges information with entities in its environment. Two properties distinguish vehicular communication from the general wireless case and drive most of its design decisions.

The first is that most safety traffic is broadcast and connectionless. A vehicle announcing its position does not know, and need not know, which vehicles are listening. There is no association, no acknowledgement and no retransmission which means that a lost packet is simply lost, and the transmitter never learns of it. The second is that the topology changes continuously: two vehicles on opposing carriageways close at a relative speed exceeding 50 m/s, and a link between them may exist for only a few seconds. Any protocol requiring a lengthy setup phase is unusable, which is why V2X safety communication is built around periodic single-hop broadcast rather than routed, connection-oriented transport.



2.2 Communication Types

Vehicle-to-Vehicle (V2V) is the direct exchange of information between vehicles. Its canonical application is the periodic broadcast of a status message the Basic Safety Message (BSM) of SAE J2735 [5] or the Cooperative Awareness Message (CAM) of ETSI [7] containing position, speed, heading, acceleration and vehicle state, at 1–10 Hz. It has the tightest latency requirement, because forward collision warning and intersection movement assist have deadlines measured in tens of milliseconds, and the least infrastructure dependency.

Vehicle-to-Infrastructure (V2I) is communication with a Roadside Unit (RSU). The RSU broadcasts Signal Phase and Timing (SPaT) and MAP messages describing the intersection, and

hazard notifications; vehicles report their presence and observations. V2I has a structural advantage for cooperative perception: an RSU can be placed where visibility is good, is not moving, and has a stable, mains-powered, network-synchronised clock and power supply.

Vehicle-to-Pedestrian (V2P) extends the safety envelope to vulnerable road users. Its difficulties are distinctive: the device is battery-constrained, its position estimate comes from a consumer-grade GNSS receiver, and it is often carried in a pocket, which attenuates the signal considerably.

Vehicle-to-Network (V2N) is communication with the cellular network and, through it, with edge servers and cloud services. It has the greatest reach and the greatest latency. Multi-access Edge Computing (MEC) exists to make network-mediated fusion possible without a round trip to a distant data centre.

Table 2.1 The four V2X communication modes.

Mode	Counterpart	Typical range	Latency class	Primary applications	Key constraint
V2V	Other vehicles	100-500 m	Lowest (tens of ms)	Collision warning, platooning, cooperative perception	Highly dynamic topology; no acknowledgements
V2I	Roadside units	100-500 m	Low (tens of ms)	SPaT/MAP, hazard warning, infrastructure-assisted perception	Requires deployed infrastructure
V2P	Pedestrian devices	50-200 m	Low-moderate	Protection of vulnerable road users	Battery life and GNSS accuracy of the device
V2N	Network / edge / cloud	Cell-sized	Highest (tens-hundreds of ms)	Traffic information, HD maps, edge-fused perception	Latency; dependence on coverage

2.3 Message Sets

Table 2.2 Standardised V2X message types.

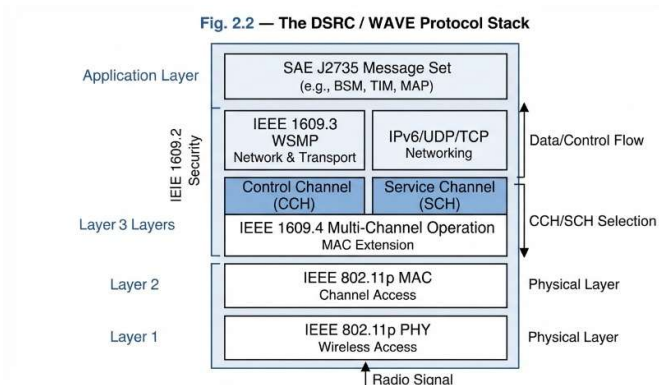
Message	Standard	Content	Typical rate
BSM	SAE J2735 [5], J2945/1 [6]	Position, speed, heading, acceleration, brake status, vehicle size	1-10 Hz
CAM	ETSI EN 302 637-2 [7]	European equivalent of the BSM	1-10 Hz
DENM	ETSI EN 302 637-3 [8]	Event-triggered hazard notification	Event-driven
SPaT / MAP	SAE J2735 [5]	Signal phase and timing; intersection geometry	~10 Hz / low rate
CPM	ETSI TS 103 324 [9]	Objects detected by the sender's sensors, with confidence and timestamps	1-10 Hz

The Collective Perception Message deserves attention: it is the standardised realisation of object-level cooperative perception, in which a vehicle shares a list of the objects it has detected rather than its own state. It is the point at which the communications standards and the cooperative perception literature meet and also where they diverge, since the standard specifies an object list while most research argues that fusing intermediate feature maps preserves considerably more information (Section 5.3). Every one of these messages carries a generation timestamp, and the whole of Section 4 rests on that field.

2.4 DSRC and IEEE 802.11p

Dedicated Short-Range Communications (DSRC) is the older of the two radio families and has the longer field-trial record [1]. It is built on **IEEE 802.11p** [2], an amendment to the Wi-Fi standard adapted to the vehicular environment.

The adaptations are targeted. Ordinary Wi-Fi requires a station to associate with an access point before transmitting; 802.11p introduces an outside-the-context-of-a-BSS mode in which a station transmits immediately, without association or handshake which is what makes it usable in a topology that changes every few seconds. The channel bandwidth is halved from 20 MHz to 10 MHz, doubling the OFDM symbol duration and hence the guard interval, a necessary concession to the long delay spreads produced by multipath reflections. Operation is at 5.9 GHz. Above the PHY and MAC sit the IEEE 1609 (WAVE) family: 1609.4 for multi-channel operation [4], 1609.3 for networking services, and 1609.2 for security [3], discussed in Section 12.



Medium access uses **CSMA/CA with EDCA**. A node wishing to transmit senses the channel; if it is busy the node defers and backs off for a random interval before retrying. This is decentralised and needs no coordinator its great virtue. Its weakness is equally structural: as the number of contending nodes grows, so does the probability of overlapping backoff intervals, and both the mean delay and its variance rise. Because safety messages are broadcast they are neither acknowledged nor retransmitted, so a collision is an irrecoverable loss.

2.5 LTE-V2X and Cellular V2X

Cellular V2X (C-V2X) was introduced by 3GPP in Release 14 [10] and offers two interfaces. The **Uu interface** is the conventional cellular uplink and downlink through a base station; it can reach any vehicle in the cell but involves at least two air-interface traversals, so its latency is structurally higher. The **PC5 sidelink** allows vehicles to transmit directly to one another without the message passing through the network. In **Mode 3** the base station schedules sidelink resources, which requires coverage; in Mode 4 vehicles select their own resources autonomously using sensing-based Semi-Persistent Scheduling (SPS) [11], with no network coverage required. **Mode 4** is the mode directly comparable with DSRC.

Sensing-based SPS works quite differently from CSMA/CA, and the difference explains most of the behavioural contrast between the two technologies. Rather than sensing the channel immediately before transmitting, a vehicle listens over a sensing window of the recent past, forms a picture of which resources in the upcoming selection window are likely to be occupied, excludes those, and selects a resource from among the remainder. Having selected one, it reserves it for a number of consecutive periods and signals the reservation in its control information so that others can avoid it.

The consequences are twofold. Because transmissions are aligned to a subframe grid and the resource has been reserved in advance, the delay experienced by a packet is bounded and its variance is small. But because the reservation is semi-persistent, two vehicles that select the same resource will keep colliding until one of them re-selects. C-V2X therefore trades the unbounded, contention-driven delay of CSMA/CA for a bounded delay with a more persistent collision structure.

2.6 NR-V2X

NR-V2X, standardised from 3GPP Release 16 onwards [12], [14], is the 5G New Radio evolution of the sidelink. It does not replace LTE-V2X so much as extend it. Its significant additions are unicast and groupcast sidelink communication alongside broadcast, which allows a receiver to acknowledge and a transmitter to retransmit a direct attack on the reliability problem created by unacknowledged broadcast; HARQ feedback on the sidelink; flexible numerology, so that the slot duration can be scaled short enough for a latency target of a few milliseconds; and much higher data rates.

The last point matters most here. The advanced use cases identified in the 3GPP study work cooperative perception, cooperative manoeuvre, remote driving, close-gap platooning have throughput and reliability requirements that IEEE 802.11p and LTE-V2X cannot meet. NR-V2X is the first radio access technology whose targets are compatible with sharing intermediate feature maps between vehicles at 10 Hz. IEEE has pursued a parallel evolution in IEEE 802.11bd, which stands in the same relation to 802.11p as NR-V2X does to LTE-V2X, and which maintains backward compatibility with deployed equipment [13].

2.7 Evolution Towards 6G

The trajectory of the last two decades from a broadcast channel designed to carry a position and a velocity, to a sidelink capable of carrying feature tensors has an obvious extrapolation, and the 6G research programme has begun to articulate it [15], [16]. Four themes recur, and Section 8 treats them in detail: sub-millisecond and deterministic latency; integrated sensing and communication, in which the radio waveform senses as well as carries; AI-native network design; and pervasive edge intelligence.

2.8 Comparative Summary

The two radio families are not competing implementations of one idea; they embody different assumptions about how a shared medium should be arbitrated. Comparative studies of their performance under varying density and PHY/MAC configuration [53]-[55] report the qualitative contrast summarised below.

Table 2.3 Comparison of DSRC/IEEE 802.11p and C-V2X (LTE-V2X Mode 4).

Aspect	DSRC / IEEE 802.11p	C-V2X (PC5 Mode 4)
Standardisation body	IEEE / SAE	3GPP
Physical layer	OFDM, 10 MHz channel	SC-FDMA, subframe-based
Medium access	CSMA/CA with EDCA (contention-based)	Sensing-based SPS (reservation-based)
Synchronisation requirement	Not required	Required - GNSS or eNB timing reference
Infrastructure dependency	None	None in Mode 4
Latency behaviour	Low at low load; grows and becomes highly variable as load rises	Bounded by the selection window; stable variance
Collision behaviour	Independent per packet	Semi-persistent - a collision can recur across periods
Reliability mechanism	None for broadcast	Blind retransmission; HARQ feedback in NR-V2X
Range	Typically up to ~300 - 500 m	Generally reported to be greater at equal transmit power [53]
Maturity	Long field-trial history	Later, with a defined path to NR-V2X and beyond

3. COMMUNICATION CHALLENGES IN V2X

3.1 Introduction

Section 2 described the V2X link as the standards intend it to work. This section describes what actually happens to a message on that link. The impairments treated here are not exotic failure modes; they are the normal operating condition of a vehicular network.

3.2 Communication Delay

Delay, or latency, is the interval between the instant an observation is made at the source and the instant it becomes usable at the destination.

$$\tau_{e2e} = t_{use} - t_{gen} = \tau_{sens} + \tau_{proc}^{tx} + \tau_{queue} + \tau_{trans} + \tau_{prop} + \tau_{proc}^{rx}$$

where the terms are, respectively, sensing and feature extraction, protocol-stack processing at the transmitter, waiting for the medium, transmission, propagation, and processing at the receiver.

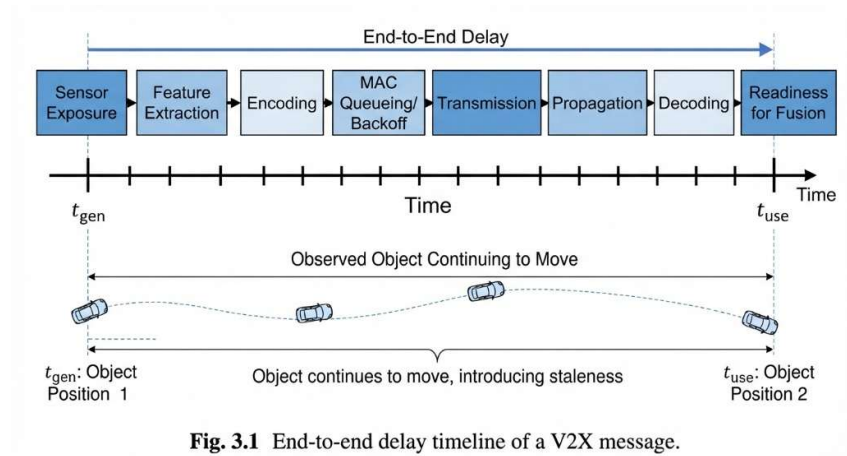


Fig. 3.1 End-to-end delay timeline of a V2X message.

Processing delay is the time taken to verify, decode and hand the packet up the stack. In V2X the dominant term is often security processing: every safety message is digitally signed under IEEE 1609.2 [3], and the receiver must verify that signature before acting. A vehicle in dense traffic may receive several hundred signed messages per second, each requiring an elliptic-curve verification. This is not merely a performance issue it is the property a denial-of-service attacker exploits(Section 12.4).

Queueing delay is the time a packet waits after it is ready to be sent but before the medium is available. It is the component with the greatest variance and the one that most directly reflects the medium-access mechanism. Under CSMA/CA a node that finds the channel busy defers, then draws a random backoff counter uniformly from a contention window CW and decrements it only while the channel is idle.

For broadcast safety messages there is no acknowledgement and therefore no detected failure, so the contention window does not grow meaning collisions are neither detected nor avoided on subsequent attempts. A useful load metric is the Channel Busy Ratio (CBR), the fraction of time the channel is sensed busy, which ITS congestion control attempts to regulate by reducing message rate or transmit power. Under sensing-based SPS the corresponding wait is bounded by the selection window, which is the structural reason for the latency-variance difference noted in Table 2.3.

Propagation delay is τ_{prop} . For a 300 m V2V link this is 1 μs four orders of magnitude below the other components. It is stated only to make the point explicit: latency in V2X is not caused by the speed of light; it is caused by contention, queueing and processing.

Transmission delay is $\tau_{\text{trans}} = L/R$, for a packet of L bits at rate R . This term scales with what is being sent, and therefore governs the feasibility of cooperative perception. A 250 kB compressed feature map at 6 Mb/s takes roughly 333 ms to transmit.

3.3 Variable Delay (Jitter)

If the delay were constant, compensation would be straightforward. The difficulty is that it varies from packet to packet.

Jitter arises almost entirely from the queueing term: a packet transmitted when the channel happens to be idle experiences almost no queueing delay, while one transmitted during a burst of activity may experience many multiples of the mean. Its consequences are that a receiver cannot assume a fixed compensation horizon but must estimate the delay per message; that messages from the same sender can arrive out of order or with irregular spacing; and that messages from different senders describing the same instant arrive at different times, so a receiver is always fusing observations whose ages differ.

3.4 Packet Loss

A packet that never arrives is worse than one that arrives late, because there is nothing to compensate. The standard metric is the **Packet Delivery Ratio**, $\text{PDR} = N_{\text{received}}/N_{\text{transmitted}}$, evaluated over a distance bin. Packets are lost for four distinct reasons, and they respond to different remedies.

Collision: Two nodes transmit on the same resource and neither packet is decodable. This is load-driven: it increases with density and message rate.

Insufficient signal-to-noise ratio: The received power falls below the receiver's sensitivity. This is distance-driven. In a log-distance path-loss model,

$$P_{\text{rx}}(d) = P_{\text{tx}} - PL(d_0) - 10 n \log_{10} \left(\frac{d}{d_0} \right) + X_{\sigma}$$

where n is the path-loss exponent and X_σ a zero-mean Gaussian shadowing term. Because X_σ is random, PDR does not fall to zero abruptly at a range boundary; it decays over a transition region.

Obstruction: a heavy vehicle between transmitter and receiver adds attenuation of the order of 10–20 dB, enough to break a link that would otherwise close. This loss is geometry-driven and highly correlated in time: it persists while the obstructing object is in the way, so losses come in bursts rather than as independent events. This distinction matters, because a scheme designed for independent losses will underestimate the length of the gaps it must bridge.

Interference, which in the intentional case is jamming (Section 12.4).

3.5 Clock Drift and Synchronisation

Every timestamp in a V2X message is written by a clock, and no two clocks agree. To first order a physical clock is an affine function of true time:

$$C_i(t) = t + \theta_i + \rho_i (t - t_0)$$

where θ_i is the offset and ρ_i the drift rate - the fractional frequency error of the oscillator, of the order of 10^{-5} to 10^{-6} for an inexpensive crystal, conventionally quoted in parts per million.

The consequence is easy to compute and easy to underestimate. Two vehicles whose clocks differ in frequency by 20 ppm accumulate a relative offset of $\Delta\theta(t) = |\rho_1 - \rho_2| t$, which reaches 20 μ s after one second, 1.2 ms after one minute, and 72 ms after one hour comparable to the entire latency budget of a safety application, produced by nothing more than the two vehicles having been switched on and left to run.

The insidious property of clock error is that it is silent. A delayed packet announces itself the receiver compares the timestamp with its own clock and sees the message is old. A packet timestamped by a drifting clock announces nothing: the receiver computes an apparent age that is simply wrong.

Three synchronisation mechanisms are available. **GNSS-disciplined time** recovers time from the satellite constellation, typically to within tens of nanoseconds; it is the primary time source for C-V2X, whose SPS mechanism cannot function without a common subframe grid. Its weakness is its dependence on the sky: in a tunnel or an urban canyon it degrades or disappears, and it is also spoofable (Section 12.2). NTP [17] estimates offset from timestamps exchanged with a server and assumes a symmetric path an assumption a vehicular network violates achieving millisecond accuracy at best. PTP (IEEE 1588) [18] achieves sub-microsecond accuracy by timestamping in hardware at the physical layer; it is the mechanism of choice within a vehicle and at roadside installations, but is far harder to apply between independently moving vehicles.

The realistic conclusion is that synchronisation is good but not perfect, and a cooperative system must tolerate the residual.

3.6 Asynchronous Vehicles

Even with perfectly synchronised clocks and zero delay, cooperative perception would still face an alignment problem, because sensors on different vehicles are not triggered at the same instant. A LiDAR spinning at 10 Hz produces a frame every 100 ms, but the phase of that rotation is arbitrary it depends on when the sensor was powered up. Two vehicles with identical 10 Hz LiDARs may capture their frames 47 ms apart indefinitely, with no clock error whatever. Different vehicles may run sensors at different rates, and a spinning LiDAR does not sample a frame instantaneously.

The total temporal misalignment between an ego observation and a received one is therefore

$$\Delta t = \underbrace{\tau_{e2e}}_{\text{communication}} + \underbrace{\Delta\theta}_{\text{clock offset}} + \underbrace{\delta_{\text{phase}}}_{\text{sampling phase}}$$

and only the first term is what is usually meant by “latency”. A system that compensates only for the first will still be misaligned by the other two.

3.7 Interruptions

An interruption is an extended period during which no packets are received from a particular neighbour. It differs from ordinary packet loss in duration and in structure: individual losses are approximately independent, whereas an interruption is a correlated burst lasting hundreds of milliseconds or seconds. Causes include a large vehicle moving between the participants, entry into a tunnel, a deep fade, congestion collapse, and cell handover.

The distinction matters because it changes what compensation must achieve. Bridging one lost packet requires extrapolating 100 ms forward; bridging a two-second interruption requires either extrapolating far beyond the horizon over which motion is predictable, or recognising that the collaborator’s contribution must be withdrawn rather than blindly predicted [31].

3.8 Bandwidth Constraints

Bandwidth constraint in V2X refers to the limited communication spectrum available for data exchange among connected vehicles and infrastructure. When many vehicles attempt to transmit simultaneously, the available bandwidth is shared, resulting in congestion and reduced network performance. This can increase latency, cause packet loss, and affect the timely delivery of safety-critical messages. Managing bandwidth efficiently is therefore essential for reliable V2X communication.

4. TEMPORAL CHALLENGES IN COOPERATIVE PERCEPTION

4.1 Why Delay Matters

A naive view holds that a 100 ms delay merely makes shared information 100 ms old, and that this is a small price for the extra field of view obtained. Table 4.1 shows this to be wrong in an important way: delayed information is not merely less useful but can be actively harmful, because a fusion module that is not told the information is stale will treat a delayed observation as a current one, and will therefore assert with confidence that an object is somewhere it is not.

The geometry is simple. An object moving at velocity v , observed at $t - \Delta t$ and treated as current at t , has a position error of $e = (v)\Delta t$; including acceleration, $e = (v)\Delta t + 1/2 (a)\Delta t^2$.

Table 4.1 Displacement error introduced by delay at typical speeds.

Object speed	50 ms	100 ms	200 ms	500 ms
30 km/h (8.3 m/s) - urban	0.42 m	0.83 m	1.67 m	4.17 m
50 km/h (13.9 m/s) - arterial	0.69 m	1.39 m	2.78 m	6.94 m
80 km/h (22.2 m/s) - highway	1.11 m	2.22 m	4.44 m	11.11 m
120 km/h (33.3 m/s) - motorway	1.67 m	3.33 m	6.67 m	16.67 m

Two observations follow. First, these errors are not small compared with the objects themselves a passenger car is about 1.8 m wide and 4.5 m long, so at highway speed and 200 ms of delay the reported bounding box may not overlap the true one at all. Second, the error is worse precisely where cooperative perception is most valuable: relative velocity is what matters for a collision, and two vehicles approaching on opposing carriageways close at the sum of their speeds.

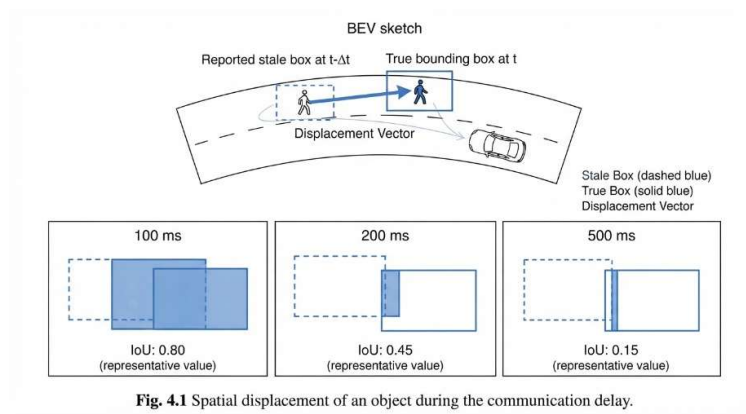


Fig. 4.1 Spatial displacement of an object during the communication delay.

4.2 Consequences for Safety

The perceptual error becomes a safety problem through three mechanisms. **False negatives in the region that matters:** cooperative perception is deployed to see what the ego vehicle cannot, and if the shared information about an occluded vehicle is stale, the estimate is wrong in exactly the region where a collision would occur. **Ghost objects:** if compensation is poor, or if fusion fails to associate a stale detection with the ego vehicle’s own current detection of the same object, the system may report two objects where there is one a failure mode identified as duplicate detection at high latency in both the empirical study [35] and the WACV 2025 work [36]. **Loss of consistency between agents:** two vehicles receiving different subsets of messages, with different delays, form different beliefs and may plan incompatibly.

The most uncomfortable conclusion, and one the literature states plainly, is that **uncompensated cooperative perception** can be worse than no cooperative perception at all. The empirical study of [35] found that cooperation can degrade otherwise-correct ego-vehicle perception, and that increased cooperative perception errors correlate with a higher frequency of driving violations.

4.3 Feature Misalignment

Most high-performing methods operate at the feature level, sharing intermediate representations produced by a neural encoder typically a Bird’s-Eye-View (BEV) feature map $F \in \mathbb{R}^{C \times H \times W}$, which assigns a C-dimensional descriptor to each cell of a spatial grid. Information about an object is not stored in one place but distributed across the cells it occupies and those around it, and when the object moves the informative pattern moves with it.

The naive correction applying the rigid transformation that maps the collaborator’s coordinate frame into the ego frame corrects for the fact that the two vehicles are in different places. It does not correct for the fact that the observations were made at different times, because the objects in the scene moved during the interval and they did not all move the same way. A rigid transform applies one motion to everything in the map, while the scene contains many objects with many different motions, plus static structure that did not move at all.

The literature therefore distinguishes two kinds of misalignment [32]: **spatial misalignment**, in which an object’s features appear in the wrong grid cells because the object has moved; and **semantic misalignment**, in which the content of the features is inconsistent with the current scene, because the object may have rotated, become occluded or become newly visible. Correcting the first without the second yields features that are in the right place but say the wrong thing.

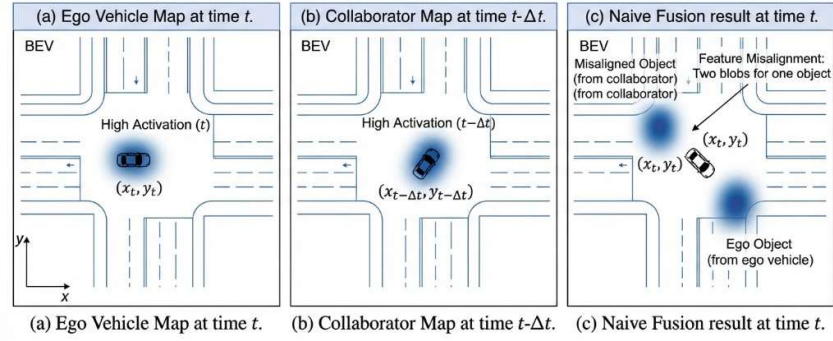


Fig. 4.2 Feature misalignment caused by asynchronous timestamps.

4.4 Temporal and Spatial Inconsistency

Temporal inconsistency takes several forms: heterogeneous ages (in a scene with three collaborators the messages may be 40, 90 and 180 ms old, so fusion must align four different instants); irregular arrival, since a method that quantises delay into a small set of discrete values will mis-compensate whenever the true delay falls between them [29]; sampling-phase offsets (Section 3.6); and interruption gaps, during which the age of the most recent information grows without bound.

Spatial inconsistency is the counterpart problem in the pose domain. What makes this awkward is the interaction: a receiver observing a discrepancy between its own detection and a collaborator's cannot immediately tell whether the cause is a pose error (the whole map is shifted) or a time error (this object has moved). CoAlign [27] addresses the pose case; FreeAlign [34] addresses both by discarding external localisation and clock signals altogether.

4.5 The Need for Compensation

Three responses are available, and the first two are insufficient. The system could wait for a synchronised frame, buffering its own data until timestamps match but this adds latency equal to the buffering interval, and deliberately delaying one's own perception to align it with someone else's is not an acceptable trade in a safety application. It could discard stale information, rejecting messages older than a threshold safe but wasteful, and the messages discarded are exactly the ones that would have described the occluded vehicle. Or it could compensate: accept stale information, estimate how stale it is, predict forward to the current instant, and fuse with a weight reflecting the uncertainty. Only the third is viable, and this is the consensus of the literature. What varies between methods is where compensation is done, what is predicted forward, and how the resulting uncertainty is handled.

4.6 Concepts in Temporal Compensation

Four concepts recur in the literature. They are described here at a conceptual level; the mechanisms that realise them are the subject of Sections 6 and 7. None of these concepts originates with this internship; all are drawn from the cited literature.

Delay-aware perception is the principle that the delay should be an explicit input to the perception system rather than an unmodelled disturbance: a message 30 ms old and one 300 ms old are not treated identically. This requires that the age be estimable (hence timestamps and clock synchronisation matter), that the compensation module accept it as a continuous variable, and that the fusion module be able to down-weight information whose age makes it unreliable [33].

Feature prediction is the task of estimating what a collaborator’s feature map would have been had it been captured at the ego vehicle’s current timestamp. Formally: given $F_i(t - \Delta t)$ and its history, produce $\hat{F}_i(t)$. This is a learning problem in a high-dimensional space, and it is what distinguishes feature-level from object-level compensation the latter predicts a handful of numbers per object, the former a tensor. FFNet [30] takes an interesting variant: the transmitter sends the feature map together with its temporal gradient, so that the receiver can extrapolate to any timestamp it needs.

Motion compensation is the more geometric approach: estimate the motion of the scene as a flow field over the BEV grid, then warp the received features along that flow. This is the mechanism of CoBEVFlow [29] and BEVSync [37]. It is attractive because it is interpretable, but its limitation is that flow describes where things went, not what they became the semantic misalignment of Section 4.3 is not addressed by warping alone.

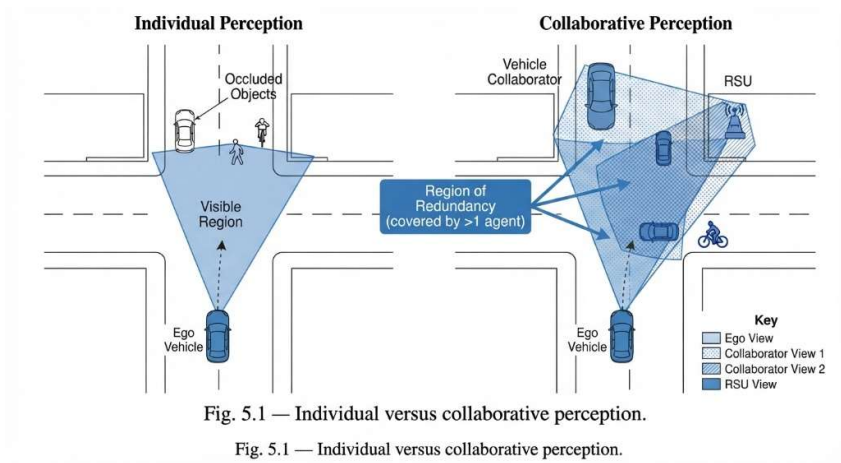
Time alignment is the prerequisite for all of the above: before compensating by Δt , the system must know Δt , and it is exactly as reliable as the clock synchronisation is. The radical alternative is to estimate the temporal offset from the data itself - matching the geometric configuration of objects observed by two agents recovers both their spatial and temporal relationship, which is what FreeAlign [34] does, with the notable side benefit of robustness to GNSS spoofing.

5. COLLABORATION IN COOPERATIVE PERCEPTION

5.1 Collaborative Intelligence

The distinction to be drawn is between individual perception, in which each agent builds its model of the world from its own sensors alone, and collaborative perception, in which agents exchange information and each builds its model from the union. The difference is not one of degree: an individual agent's model is bounded by what its sensors can physically observe, and no amount of computation recovers information that never reached the sensor. A collaborative agent's model is bounded by what the network of agents can observe, which is a strictly larger set [20].

Two distinct benefits follow, and they should not be conflated: coverage seeing what one could not see and confirmation being more certain about what one did see, because an uncertain detection is corroborated by another agent.



5.2 Why Collaboration is Necessary

Five limitations of single-agent perception motivate collaboration, and they compound.

Occlusion is the primary motivation and the one no improvement in sensor technology can address: a LiDAR return tells you about the first surface the beam strikes and nothing about what lies behind it. In dense urban traffic a large fraction of the relevant scene is behind something else.

Limited sensor range. Detection range degrades with distance in a way that is fundamental a distant object returns few LiDAR points and subtends few pixels. At highway speed, the distance over which a vehicle must anticipate a stationary obstacle exceeds the range at which any sensor gives a confident detection.

Blind spots. Every sensor configuration has regions of poor coverage. A vehicle in another's blind spot is, almost by definition, well placed to see itself.

Adverse weather. Fog scatters LiDAR returns and reduces camera contrast; rain produces spurious returns; snow covers markings. These conditions degrade sensors by range, so collaboration converts a long-range problem into a set of short-range ones: if every vehicle can see 20 m and vehicles are 20 m apart, the chain of observations still covers the road.

Dynamic traffic. A vehicle that has just braked hard knows it has braked hard and can say so, in a message that arrives long before the deceleration becomes visually apparent several places back in the queue. This is the strongest form of the argument: some information is not merely hard for a remote sensor to obtain, it is impossible.

5.3 Levels of Collaboration

The question then becomes what agents should share. Four levels are distinguished, by how much processing the sender performs before transmitting. The choice is fundamentally a bandwidth–accuracy trade-off, and it interacts with delay: the more raw the shared data, the larger it is, and by $\tau_{\text{trans}} = L/R$ - the longer it takes to send, so the staler it is on arrival.

Raw sensor sharing (early fusion). The sender transmits its point cloud or image. No information is lost, so this is in principle the upper bound on achievable accuracy but the data volume is prohibitive (a LiDAR sweep is several megabytes), and it is unusually sensitive to pose and time errors, since misaligned raw points produce a badly corrupted merged cloud. It is an important research baseline, not a deployable scheme.

Feature-level collaboration (intermediate fusion). The sender runs its raw data through the encoder stage of its network, obtains a BEV feature map, compresses it and transmits that. The encoder has already discarded most of the redundancy while retaining task-relevant information, so the representation is far smaller than raw data and far richer than an object list. This is the level at which the great majority of high-performing methods operate [21], [23], [26]. Its costs are that the representation is still large by V2X standards, and that it is model-specific two vehicles with different perception networks produce feature maps that are not directly comparable, which is a serious obstacle to interoperability between manufacturers.

Object-level collaboration (late fusion). The sender runs its full pipeline and transmits a list of detected objects position, dimensions, orientation, velocity, class, confidence. This is what the ETSI Collective Perception Message standardises [9]. The messages are tiny and interoperable, and compensation is easy, being a kinematics problem rather than a learning problem. But information is irrecoverably lost: if the sender's detector missed an object because it saw only three LiDAR points on it, the receiver never learns those points existed. Feature-level fusion can combine weak evidence from two agents into a confident detection; object-level fusion cannot, because the weak evidence was thresholded away before

transmission. This is the fundamental reason the research community favours intermediate fusion despite its cost.

Decision-level collaboration. Agents share not observations but intentions “I am changing into the left lane in 2 seconds”. This conveys information present in no sensor stream, in a message of a few tens of bytes, but it presupposes trust, and it is the level at which a malicious participant can do the most damage with the least effort.

Table 5.1 Comparison of collaboration levels.

Level	What is shared	Typical size	Bandwidth cost	Information retained	Compensation difficulty
Raw (early fusion)	Point clouds, images	MB	Prohibitive	Complete	Moderate
Feature (intermediate)	Encoded BEV feature maps	tens—hundreds of kB	High	High	Hard (tensor prediction)
Object (late fusion)	Detected object lists (CPM)	hundreds of bytes	Low	Low - pre-thresholded	Easy (kinematic extrapolation)
Decision	Intentions, manoeuvre requests	tens of bytes	Very low	Not perceptual	Not applicable

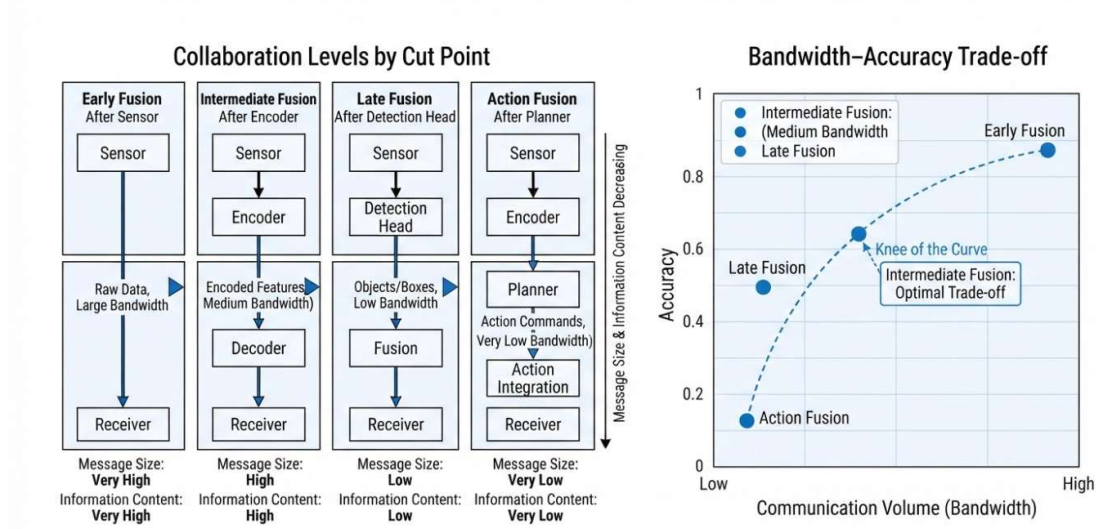


Fig. 5.2 — Levels of collaboration and the bandwidth-accuracy trade-off.

5.4 Challenges in Collaborative Perception

Communication overhead. The arithmetic of Section 3.8 is unforgiving, and the response in the literature has been selective communication sharing only the spatially critical parts of the

feature map. Where2comm [26] computes a spatial confidence map and transmits only regions where the sender’s perception is confident.

Latency and packet loss. Treated in Sections 3 and 4. One consequence specific to feature sharing is worth noting: because feature maps are large they may be fragmented across several transport packets, and the loss of any fragment destroys the whole map so the effective loss rate for feature-level sharing is considerably higher than the per-packet loss rate.

Clock drift and sensor asynchrony. Their consequence for collaboration is that the receiver’s estimate of Δt the input to every compensation module is itself uncertain.

Heterogeneity. Different vehicles carry different sensors and run different networks. Feature-level collaboration between a LiDAR-based encoder and a camera-based one is not straightforward, and the empirical study of [35] found LiDAR-based cooperation configurations to perform best among those tested.

Trust and security. A collaborating agent may be malicious (Section 12).

5.5 Applications

Intersection safety is the canonical application: a warning of a vehicle approaching on a conflicting path that is not yet visible. Cooperative perception generalises this from connected vehicles announcing themselves to any object detected by any participant which matters, because the pedestrian stepping off the kerb is not carrying an on-board unit.

Lane-change assistance addresses the blind-spot problem of Section 5.2. **Cooperative Adaptive Cruise Control** receives the acceleration command of the vehicle ahead rather than merely observing its deceleration, allowing vehicles in a platoon to react simultaneously rather than sequentially. **Autonomous driving** consumes the extended and confirmed scene directly. **Emergency vehicle awareness** lets an ambulance announce its position and route far beyond the range at which its siren is audible inside a modern insulated cabin.

5.6 Towards an Architecture

This section has established that collaboration is necessary because single-agent perception is limited by geometry and physics; that it is most valuable at the feature level, where the accuracy of early fusion is approached at a fraction of its bandwidth cost; and that collaboration at that level is exactly where the temporal problems of Sections 3 and 4 bite hardest, because feature maps are large hence slow to transmit, hence stale and hard to predict forward. The system that resolves this tension needs a definite structure, which Sections 6 and 7 now supply.

6. MACHINE LEARNING FOR FEATURE PREDICTION

6.1 Why Machine Learning is Used

The compensation problem is: given a collaborator’s feature map captured at $t - \Delta t$ estimate what it would have contained at t . Two routes exist.

The **analytical route** models the scene explicitly detect objects, estimate velocities, extrapolate positions with a kinematic model, re-render the features. It is transparent and cheap, and it fails wherever the model fails: a constant-velocity model cannot anticipate braking, and no kinematic model can express what happens to the appearance of an object as it rotates relative to a sensor or slides behind an occluder.

The **learned route** treats the mapping $F(t - \Delta t) \mapsto \hat{F}(t)$ as a function to be estimated from data, on the justification that it depends on regularities how vehicles actually move, how scenes actually evolve that are present in large datasets but hard to write down in closed form. This is the route the literature almost uniformly follows, with the instructive exception of CoDynTrust [33], which deliberately uses simple linear extrapolation and invests its modelling effort in uncertainty rather than in prediction, on the reasoning that knowing how much to trust a prediction may matter as much as making it more accurate.

6.2 The Prediction Tasks

Four related tasks appear under this heading, and conflating them causes confusion. **Motion prediction** estimates the future trajectory of an object a few numbers per object. Future state estimation does the same for the whole scene. **Feature prediction** estimates the future value of the representation, a $C \times H \times W$ tensor. **Temporal prediction** is the general operation of estimating a signal at a timestamp for which no sample exists and it is worth noting that compensation is strictly extrapolation, not interpolation, which is the harder of the two.

The distinction the field has converged on is between predicting in the feature space and predicting in the object space. Object-space prediction is easy and interpretable but discards everything the encoder learned; feature-space prediction preserves the representation but requires transforming a high-dimensional tensor. The compromise most successful methods adopt is to predict motion in a low-dimensional space a flow field, a set of region-of-interest velocities and use that motion to transform the high-dimensional features, which is what CoBEVFlow [29] and the flow-prediction module of LRCP [36] do.

6.3 Convolutional Neural Networks

Their two relevant properties are weight sharing the same kernel is applied everywhere, so the parameter count does not grow with grid size and translation equivariance. Both are

exactly what a BEV grid requires. In cooperative perception, CNNs appear as the encoder that turns a point cloud into a BEV feature map (via PointPillars [51] or VoxelNet [52] style voxelisation), as a fusion operator, and as the warping operator in flow-based compensation. What they cannot do alone is model time: a CNN applied to a single frame has no memory, and stacking frames along the channel dimension fixes the temporal window and treats time as just another spatial axis a poor inductive bias when frames are irregularly spaced, which as Section 3 established they always are.

6.4 Transformers and Attention

Three properties make this unusually well suited to cooperative perception, and explain why the most recent methods are built on it.

It **handles a variable number of inputs natively** the number of collaborators changes as vehicles come into and go out of range, and an attention layer over a set of collaborator features does not care how many there are, whereas a fixed-size CNN does. It **weights inputs by relevance rather than position**, which is precisely what a delay-aware fusion module needs: a feature from a collaborator 300 ms stale with a poor view should contribute less than one 40 ms old with a clear view, and V2X-ViT [23] builds its entire fusion module on this principle. And it can **attend** across time as well as across space the property TraF-Align [32] exploits, generating temporally ordered sampling points along each object’s predicted trajectory and directing attention from a current-time query to the historical features lying along that path. The query asks what was at each point at each earlier time, and attention gathers the answer, producing a reconstruction informed by the whole trajectory rather than a single warped snapshot.

A related idea, deformable attention, restricts each query to a small set of learned sampling offsets rather than the entire feature map, reducing the quadratic cost of full attention to something tractable on a dense BEV grid. It is the mechanism on which LRCP [36] builds its asynchronous fusion.

6.5 Two Cautions

Prediction has a horizon. Beyond a few hundred milliseconds, vehicle motion becomes genuinely unpredictable a driver may brake or turn, and no model, however large, recovers information that does not exist. Compensation extends the usable range of delay; it does not abolish it. This is why the reviewed papers evaluate over a bounded latency range, typically up to 400–500 ms.

A confident wrong prediction is worse than an uncertain one. A module that outputs a sharp, high-confidence feature map for an object that has in fact braked hard will cause fusion to assert the object’s presence where it is not. This is the argument for coupling prediction with uncertainty estimation, and it is the explicit design principle of CoDynTrust [33].

7. COOPERATIVE PERCEPTION ARCHITECTURE

7.1 The Pipeline

The material of the preceding sections assembles into a five-layer pipeline. The decomposition is conceptual, adopted for clarity; in a real implementation the boundaries are not always crisp, and in end-to-end trained systems several layers are learned jointly.

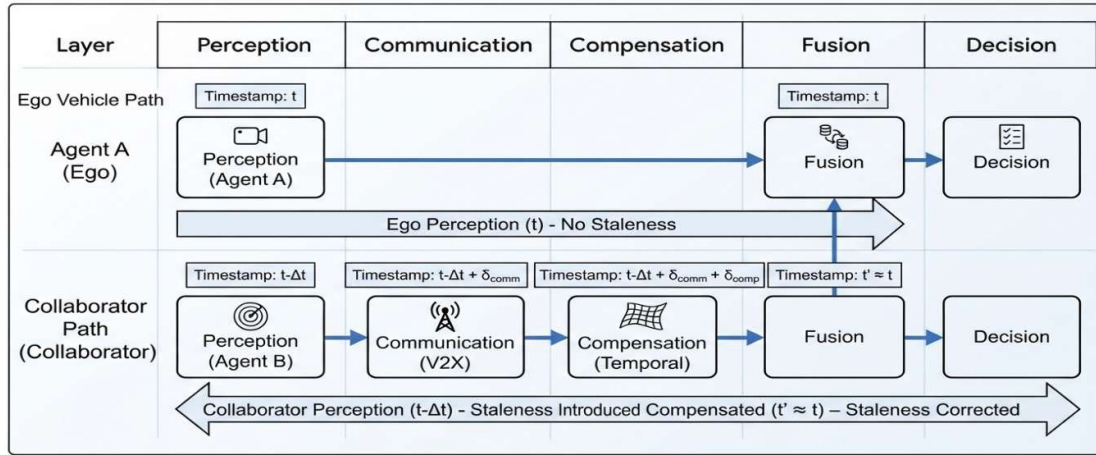


Fig. 7.1 The five-layer cooperative perception pipeline

Layer	Input	Output	Responsibility
Perception	Raw sensor data	BEV feature map or object list, with a timestamp	Encode the local observation
Communication	Encoded representation	Received messages with timestamps and delay estimates	Select, compress, transmit; handle loss
Compensation	Delayed features + Δt	Predicted current-time features + confidence	Correct for elapsed time and pose error
Fusion	Ego features + compensated collaborator features	Unified scene representation	Combine, weighting by reliability
Decision	Unified representation	Detections, tracks, plans	Detect, track, predict, plan

7.2 Perception and Communication Layers

The **perception layer** converts raw sensor data into a shareable representation. For LiDAR the standard approach is voxelisation followed by a convolutional encoder [51], [52]; for cameras the image must first be lifted into the BEV plane. Three properties of its output matter

downstream. The representation must carry a timestamp and the timestamp must correspond to the instant the sensor observed, not the instant the feature was computed, which for a spinning LiDAR differ by up to a full rotation period. It must be compressible (Section 3.8). And it should ideally be interoperable, which in practice it is not.

The **communication layer** has three perception-specific responsibilities. Selection: given a bandwidth budget, which parts of the feature map should be transmitted? The literature’s answer is the parts the receiver most needs and least already knows [26]. Compression: heavier compression reduces τ_{trans} and hence staleness but degrades information content, a direct trade-off. Delay estimation: the receiver must estimate Δt for each message, and this estimate is only as good as the clock synchronisation.

7.3 The Compensation Layer

This layer distinguishes a system designed for real communication conditions from one designed for the idealised synchronous assumption. It has five functions.

Delay compensation is the core function. Three families of mechanism appear in the literature, and the distinction structures the field. Warp-based methods estimate a motion field over the BEV grid and displace features along it [29]. Transmitter-side prediction has the sender transmit a representation that is intrinsically extrapolable FFNet [30] sends the feature together with its temporal gradient, so the receiver can evaluate it at whatever timestamp it needs. Attention-based reconstruction rebuilds the current-time feature by attending to historical features along an object’s trajectory [32], and is best able to address semantic as well as spatial misalignment. Earlier latency-aware work [28] established the general principle that shared features must be actively synchronised before fusion.

Packet-loss and interruption handling. When a message is lost, there is nothing to compensate. The naive response reuse the last received map amounts to compensating over a longer Δt , acceptable for one lost packet but not for a sustained interruption, because the prediction horizon of Section 6.5 is exceeded. The considered response is to maintain a history buffer for each collaborator, use it to reconstruct what can be reconstructed, and degrade gracefully towards single-agent perception as the interruption lengthens. This is the design of V2X-INCOP [31].

Feature alignment transforms the collaborator’s features into the ego frame, and under pose error this transformation is itself wrong. CoAlign [27] enforces consistency among agents’ pose estimates; FreeAlign [34] discards the external pose source entirely, matching graphs of salient detected objects to recover relative pose and time from geometry.

Prediction is the module of Section 6, producing $F_i(t)$ from $F_i(t-\Delta t)$ and its history.

Confidence estimation. The output should be a feature map and a statement of how much to trust it, for the reason given in Section 6.5. Confidence should depend on the magnitude of Δt , on the predictability of the region, on the uncertainty of the underlying detection, and on

the reliability of the collaborator which is where the security material of Section 12 re-enters the architecture. CoDynTrust [33] operationalises this by generating a dynamic feature trust modulus per region of interest, built from modelled aleatoric and epistemic uncertainty, and allowing that uncertainty to propagate onward to planning and control rather than being discarded at the perception boundary.

7.4 Fusion and Decision Layers

Fusion combines the ego vehicle’s features with the compensated features of all collaborators. The design space runs from element-wise maximum or mean over aligned maps cheap, permutation-invariant and surprisingly hard to beat as a baseline through learned convolutional fusion, which is sensitive to the number of agents, and graph-based fusion [21], to attention-based fusion [23], which weights each agent’s contribution at each location by learned relevance. Attention is dominant in current work, and it composes most naturally with the confidence estimates of Section 7.3, since the confidence can simply be injected as a bias into the attention weights.

The **decision layer** feeds detection, tracking, prediction and planning. Two points connect it back to the theme of this report. First, uncertainty should not be discarded at the perception boundary: a planner that knows a detection is uncertain can behave conservatively, while one given a point estimate cannot [33]. Second, perception errors have behavioural consequences: the empirical study of [35] evaluated cooperative perception online over test routes and found that increased perception errors were associated with a higher frequency of driving violations, and that the systems tested were not robust to communication interference when running online. This is a strong argument for evaluating cooperative perception in closed loop, on driving outcomes, and not only on average precision computed offline.

8. EVOLUTION FROM 5G TO 6G

8.1 Limitations of 5G for Cooperative Perception

5G, and specifically NR-V2X, is a substantial improvement over what preceded it, and it nonetheless falls short of what cooperative perception requires, in four respects.

Latency. The 5G URLLC target of approximately 1 ms is a radio-interface figure the time to get a packet across the air interface not an end-to-end figure. The application-level latency a perception system experiences includes the sensing, encoding, queueing and decoding terms of Section 3.2, and in the V2N case the path to and from an edge server. The end-to-end figure is an order of magnitude larger than the headline number.

Reliability. The URLLC reliability target is defined for particular packet sizes and conditions. Sustaining it for large, frequent feature-map transmissions among a dense population of vehicles contending for the same sidelink resources is a different problem.

Scalability. The failure mode of a congested channel is mitigated but not solved by the reservation-based sidelink; the arithmetic of Section 3.8 is unchanged, and the number of vehicles in a traffic jam is not small.

Absence of native intelligence. In a 5G network the perception model and the network are separate: the model produces bits, the network carries them, and neither knows anything about the other. The network cannot prioritise the feature-map region containing an occluded pedestrian over the region containing empty road, because it has no idea which is which.

8.2 What 6G Proposes

Sub-millisecond, deterministic latency. The interesting target is not merely a lower mean but a bounded tail: a safety-critical control loop cares about the 99.999th percentile, not the average, and it is the tail that governs whether the loop is stable.

Integrated sensing and communication (ISAC). The same waveform carries data and senses the environment through the reflections of the transmitted signal. The implication for cooperative perception is direct: an ISAC-enabled RSU is itself a sensor, contributing observations without additional sensor hardware and, because sensing and communication share a clock, contributing them with a timestamp intrinsically aligned to the communication timebase. This attacks the synchronisation problem of Section 3.5 at its root.

AI-native networking. Learning becomes a component of the network rather than an application on top of it. Two consequences matter. Semantic communication: rather than transmitting bits faithfully and letting the application interpret them, transmit only what is semantically necessary for the receiver's task the selection problem of Section 7.2, moved from the application into the network. Predictive resource allocation: a network that can

anticipate where vehicles will be can schedule resources before they are requested, removing the request–grant round trip from the latency budget.

Pervasive edge intelligence. Computation close enough to the vehicle that a fused scene can be computed centrally and returned within the safety budget. Central fusion eliminates the pairwise exchange cost of the fully distributed case, at the price of a single point of failure and of attack.

Table 8.1 Comparison of 5G and 6G target requirements.

Requirement	5G (NR-V2X)	6G (research targets)	Relevance to cooperative perception
User-plane latency	~1 ms radio interface; tens of ms end-to-end	Sub-millisecond; bounded tail	Sets the compensation horizon Δt
Reliability	99.999% (URLLC, specified conditions)	Up to 99.99999% targeted	Determines how often compensation must bridge a gap
Peak data rate	~20 Gb/s (ideal)	~1 Tb/s targeted	Determines whether raw or feature-level sharing is feasible
Connection density	~10 ⁶ devices/km ²	~10 ⁷ devices/km ² targeted	Determines behaviour in congested traffic
Positioning accuracy	~1 m typical	Centimetre-level targeted	Reduces the pose error of Section 4.4
Synchronisation	GNSS / network-based, μ s-level	Tighter, GNSS-independent options	Reduces the clock term of Section 3.5
Sensing	Not integrated	Integrated (ISAC)	New observation source with an aligned timebase
Intelligence	Applications only	AI-native; semantic communication	Moves feature selection into the network

8.3 Two Cautions

The 6G literature is written in an optimistic register, and two cautions should be recorded. First, the compensation problem does not disappear: even with zero communication delay, the sampling-phase offset of Section 3.6 remains, because sensors on different vehicles are still triggered independently. Temporal compensation is required by the sensors, not only by the network. Second, capability creates demand: every previous increase in vehicular communication capacity has been met by an increase in what the perception community wishes to share from a position and a velocity, to an object list, to a compressed feature map, to multi-frame feature tensors. The constraint of Section 3.8 will still bind; it will simply bind at a different point.

9. SIMULATION FRAMEWORKS STUDIED

9.1 Scope

The frameworks in this section were studied, few of them were used. Their architecture, capabilities and limitations were examined through their documentation and reference publications, in order to understand what tooling the field relies on and in particular what each tool does and does not model, which is a prerequisite for reading the experimental sections of research papers critically. They were not used to produce any result reported in this report.

The central observation is that no single tool models the whole problem. V2X research needs realistic vehicle mobility, a realistic wireless channel and protocol stack, and realistic sensor data. Traffic simulators provide the first, network simulators the second, and driving simulators the third; the tools that matter most are those that couple them.

9.2 SUMO

SUMO [48] is an open-source microscopic traffic simulator: it models each vehicle individually, with its own car-following model, lane-changing model and route, and produces the trajectory of every vehicle at every timestep. Its role in V2X research is to generate realistic mobility the positions a network simulator needs as input must come from somewhere, and randomly generated mobility produces traffic that does not behave like traffic, with no queues, no platoons and no signal-driven bunching. Since the behaviour of a V2X network depends entirely on how many vehicles are in range of one another, this matters a great deal.

Advantages: mature; scales to large networks; imports real road geometry from OpenStreetMap; exposes a control interface (TraCI) allowing an external program to read and modify the simulation state at each timestep. *Limitations:* it models mobility only no radio, no protocol stack, no sensors.

9.3 OMNeT++ and Veins

OMNeT++ [47] is a discrete-event simulation framework a general-purpose engine rather than a network simulator as such on which the INET framework and Veins are built. Its architecture is clean and modular and its visualisation excellent, which makes it well suited to teaching; the learning curve is steep, and the fidelity of any simulation depends entirely on the fidelity of the models chosen.

Veins [46] couples OMNeT++ and SUMO and is the standard tool for IEEE 802.11p simulation. Its defining property is bidirectional coupling: SUMO advances the traffic simulation and reports vehicle positions to OMNeT++ over TraCI; OMNeT++ simulates the radio communication among vehicles at those positions; and critically OMNeT++ can then instruct SUMO to change a vehicle's behaviour in response to a message received. Without this

feedback path one could not simulate an application in which a warning causes a vehicle to brake, which is to say one could not simulate most of what V2X is for.

Advantages: a well-validated IEEE 802.11p PHY and MAC model, including the 10 MHz channel, EDCA and obstacle-based shadowing; a large body of published work using it, which makes results comparable. *Limitations:* its C-V2X support is less mature than its 802.11p support; it does not model sensors, so it can simulate cooperative awareness but not cooperative perception; and detailed PHY evaluation makes large-scale simulation computationally heavy.

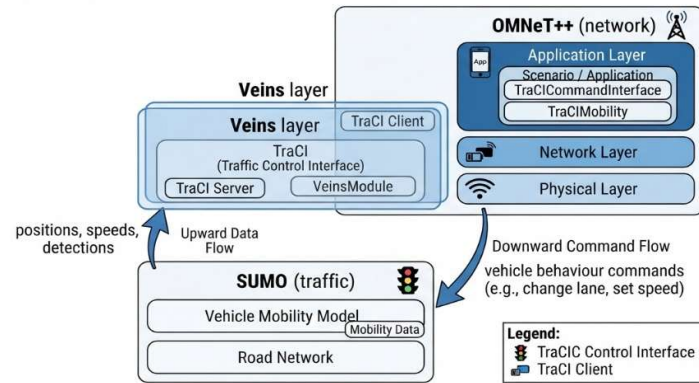


Fig. 9.1 Veins architecture: SUMO + OMNeT++ integration via TraCI.

9.4 ns-3

ns-3 [50] is a discrete-event network simulator with a strong reputation for the fidelity of its physical-layer and protocol models. Its LTE and NR modules, and the C-V2X sidelink models built on them, make it the natural counterpart to Veins for the cellular side. *Advantages:* detailed, well-validated PHY models; active development including 5G NR; can be coupled to SUMO for mobility. *Limitations:* like OMNeT++, it models the network and not the sensors; steep learning curve; expensive at scale.

9.5 CARLA

CARLA [49] is an open-source driving simulator built on a game engine. It renders a 3D world and simulates sensors within it cameras, LiDAR, radar, GNSS, IMU producing synthetic data with perfect ground truth. It is the source of the sensor data a cooperative perception system consumes: every synthetic dataset in Section 10 was generated in CARLA, and it is the platform on which closed-loop evaluation is done, including the empirical study of [35].

Advantages: photorealistic rendering; a rich, configurable sensor suite; perfect ground-truth annotation, which would be impossibly expensive to obtain in the real world; deterministic and repeatable; scriptable; and capable of generating data with controlled delays and asynchrony, which is what makes it possible to evaluate a compensation method at all. *Limitations:* it has only a rudimentary communication model, so delays must be injected

artificially rather than emerging from a simulated protocol stack; it suffers from the sim-to-real gap; and it is GPU-demanding.

9.6 OpenCOOD

OpenCOOD [22] is not a simulator but an open-source codebase and benchmark for cooperative perception, providing dataloaders for the standard datasets, implementations of the standard fusion strategies and of a number of published models, and a common evaluation protocol. It is the reason results in this field are comparable at all: because a large fraction of published methods are implemented against it. TraF-Align [32] and LRCP [36] both state they are built on it a new method can be evaluated against strong baselines on identical data.

Advantages: strong baselines; standardised evaluation; and direct support for injecting delay and pose error, which is precisely the experimental facility the compensation literature requires. *Limitations:* it is oriented towards offline, open-loop evaluation, measuring average precision on recorded data rather than driving outcomes; and its communication model is an abstraction a delay parameter, a loss parameter rather than a simulated protocol stack. It can therefore tell you how a method responds to a 200 ms delay, but not how often a 200 ms delay would actually occur.

9.7 How They Fit Together

The tools compose along the axis of what they model. SUMO supplies mobility; Veins or ns-3 turns that mobility into a realistic pattern of message delivery, delay and loss; CARLA turns it into sensor data; OpenCOOD turns sensor data into fused detections and measures their quality.

The gap in this chain is that the perception tools take the delay as a parameter while the network tools produce it as a result, and the two are rarely joined. A fully co-simulated pipeline in which the delay a compensation module must handle is the delay the protocol stack actually produced, under the traffic density SUMO actually generated is the natural next step, and it is identified as such in Section 15.

10. PUBLIC DATASETS

10.1 Scope

The datasets below were studied through their accompanying publications and documentation, in order to understand how cooperative perception methods are evaluated and what that evaluation does and does not capture. No model was trained on them. Two questions guided the study: what does each dataset contain, and the discriminating question how does each represent time?

10.2 OPV2V

OPV2V [22] was the first large-scale open benchmark for V2V cooperative perception. It is synthetic, generated in CARLA across a number of towns and scenarios, with 2 to 7 connected vehicles present in each frame. Each connected vehicle carries a 64-channel LiDAR, four RGB cameras giving 360° coverage, GNSS and IMU. Annotations are 3D bounding boxes with perfect ground truth the decisive advantage of a synthetic dataset, since annotating occluded objects in real data is essentially impossible, the annotator being unable to see them either, which is precisely the point of cooperative perception.

Its frames are captured synchronously, and this cuts both ways. OPV2V cannot be used to study natural asynchrony, because there is none. But precisely because it is perfectly synchronous, it is an excellent platform on which to inject controlled delay and pose error and measure the resulting degradation which is how the compensation literature actually uses it. Its limitations are the sim-to-real gap, the absence of infrastructure agents, and the fact that all communication imperfections must be simulated on top of it.

10.3 V2XSet

V2XSet [23] was introduced alongside V2X-ViT and extends the OPV2V idea in the direction that matters here: it is a synthetic CARLA-based dataset that includes both vehicles and infrastructure as agents and explicitly models communication imperfections, incorporating localisation error and transmission delay into the dataset itself rather than leaving them to be injected afterwards. This makes it the natural benchmark for methods claiming robustness to imperfect communication. Its limitation is that it remains synthetic, and that the modelled imperfections while a great improvement over assuming none are parametric abstractions rather than the output of a simulated protocol stack.

10.4 DAIR-V2X

DAIR-V2X [24] is the important counterweight: a real-world dataset collected on public roads in Beijing, covering vehicle–infrastructure cooperation, with an instrumented vehicle carrying LiDAR and camera and roadside units mounted at intersections at elevated positions.

Annotations are 3D bounding boxes, and the DAIR-V2X-C subset provides cooperative annotations fusing both viewpoints.

The key point is temporal. Because it is real, it contains real asynchrony: the vehicle’s sensors and the infrastructure’s sensors were not triggered at the same instant, the timestamps genuinely differ, and there is a real latency between capture and availability. This is exactly what a synthetic dataset cannot provide by construction, and it is why FFNet [30], BEVSync [37], TraF-Align [32] and LRCP [36] all evaluate on DAIR-V2X or on its sequential extension DAIR-V2X-Seq (V2X-Seq [25]), which adds sequential frames and is therefore the dataset of choice for methods that need history in order to predict. Its limitations are its smaller scale, its V2I focus, and the fact that annotation of occluded objects is limited by what the sensors could actually see.

10.5 What the Datasets Reveal

Three observations emerged from this study. The synthetic datasets are synchronous, and that is a methodological trap: a method evaluated only on OPV2V without injected delay is being evaluated under an assumption no deployed system will satisfy, and reported accuracy under that assumption is an upper bound rather than a performance estimate. Injected delay is not the same as real delay: drawing a delay from a small set of discrete values is a useful approximation, but the real distribution is continuous, heavy-tailed, correlated with vehicle density, and accompanied by correlated bursts of loss which is why CoBEVFlow [29] emphasises irregular continuous timestamps and LRCP [36] encodes latency as a continuous input. And real datasets are the only place asynchrony can be observed rather than assumed, which is why the most credible robustness results in the literature are reported on DAIR-V2X, and why so many papers evaluate on both a synthetic and a real dataset the synthetic one allowing a controlled sweep over delay, the real one testing whether the method survives contact with reality.

11. A comparative exercise: DSRC and C-V2X

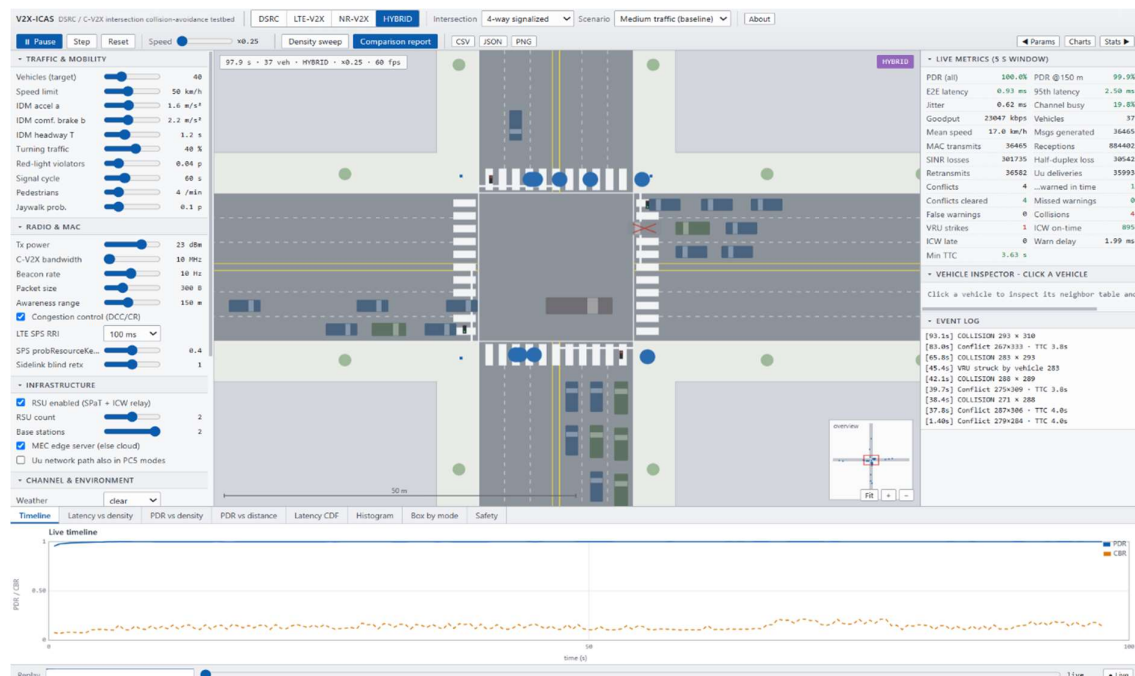
11.1 Purpose and status

This section reports a learning exercise rather than an experimental study. Having read the descriptions of CSMA/CA and of sensing-based semi-persistent scheduling in Section 2, an interactive simulator was written to represent the competing access mechanisms of DSRC and C-V2X in abstracted form, so as to observe how latency, packet delivery and for the collision-avoidance case specifically warning timeliness respond to vehicle density, and to confirm that the mechanisms had been correctly understood.

It is not a standards-compliant simulator. The physical layer is abstracted to a logistic mapping from signal-to-interference-plus-noise ratio to packet-error probability rather than a full modulation, coding and BER chain; the resource-sensing procedure of SPS is represented in reduced form; parameters were chosen for plausibility and cross-checked against published ranges rather than fitted to measurement. The intended output is the ordering and the shape of the curves which technology degrades in which manner as load rise not the absolute values

That coupling was necessary rather than decorative: the value of vehicular communication at an intersection lies precisely in carrying information the driver cannot see around the corner, so the non-line-of-sight (NLOS) geometry created by the corner buildings is the phenomenon under study, not a detail to be idealised away.

11.2 Setup



A four-way signalized intersection was modelled, with dedicated turn lanes, pedestrian crosswalks and corner buildings that occlude line of sight. Vehicles follow the Intelligent Driver Model for car-following and MOBIL for lane changes, obey the signal phases, and each broadcasts a fixed-size safety message (CAM) at a fixed rate. Propagation combines a log-distance line-of-sight path-loss law with an additional corner-NLOS excess loss, log-normal shadowing evolved as an AR(1) process decorrelated over distance, Nakagami-m fast fading, and vehicle-body blockage between transmitter and receiver.

Four access mechanisms are represented over this common channel. DSRC medium access is modelled as CSMA/CA for unacknowledged broadcast frames with energy-detect carrier sensing, per-receiver SINR against all overlapping transmitters (so hidden terminals and the capture effect appear), half-duplex reception loss, and reactive decentralized congestion control. LTE-V2X Mode 4 is modelled as a subframe-based resource grid with sensing-based semi-persistent reservation: a vehicle senses, excludes occupied resources, selects one, holds it for a reselection counter, and re-selects on expiry. NR-V2X Mode 2 adds dynamic per-packet selection with blind/HARQ retransmission. A Hybrid PC5+Uu path adds an uplink–edge/cloud–downlink network route with a handover interruption. A cooperative safety application generates collision, emergency-brake and vulnerable-road-user warnings whose delivery is checked against a braking deadline, and a separate ground-truth oracle which sees the true kinematics rather than the received messages scores missed and false warnings. Clocks are assumed perfect and security processing is not modelled; both simplifications make the reported latencies, if anything, optimistic.

Table 11.1 Setup parameters. NR-V2X Mode 2 and the Hybrid PC5+Uu path share the same channel model and are reported alongside in Table 11.2.

Parameter	DSRC (IEEE 802.11p)	C-V2X (LTE-V2X Mode 4)
Carrier / bandwidth	5.9 GHz / 10 MHz	5.9 GHz / 10 MHz
Transmit power	23 dBm	23 dBm
Message size / rate	300 B / 10 Hz	300 B / 10 Hz
Medium access	CSMA/CA broadcast, energy-detect CS, ETSI DCC	Sensing-based SPS, 1 ms subframe, reselection counter
Path loss / shadowing	$22.7 \cdot \log_{10} d + 41$ dB + corner NLOS / $\sigma = 3\text{--}4$ dB (AR-1)	$22.7 \cdot \log_{10} d + 41$ dB + corner NLOS / $\sigma = 3\text{--}4$ dB (AR-1)
Fast fading / blockage	Nakagami-m (3/2/1) / NLOSv vehicle blockage	Nakagami-m (3/2/1) / NLOSv vehicle blockage
Densities swept	10, 25, 40, 60, 85, 110	same

Table 11.2 Figures below the intersection's awareness range should be read from the 150 m distance bin, which is the safety-relevant range at an urban intersection.

RAT	Mean latency	95th-pct latency	PDR @150 m	CBR	Warn delay
DSRC (802.11p)	[0.71 ms]	[1.20 ms]	[69.1%]	[18.9%]	[0.58 ms]
LTE-V2X Mode 4	[61.0 ms]	[95.0 ms]	[69.5%]	[0.1%]	[22.9 ms]
NR-V2X Mode 2	[0.74 ms]	[2.50 ms]	[80.0%]	[8.8%]	[1.16 ms]
Hybrid PC5+Uu	[1.66 ms]	[11.0 ms]	[99.8%]	[7.3%]	[2.40 ms]

11.3 Observations

The mechanisms of Section 3 predict a clear contrast, and the exercise was constructed to see whether it appeared. Under CSMA/CA, latency should grow with density, because a node that finds the channel busy must defer and back off, and the probability of finding it busy rises with the number of contending nodes; the variance should grow faster than the mean, because backoff is a random variable whose spread widens as deferrals compound. Under sensing-based SPS the prediction differs: because a resource is reserved in advance, the wait for it is bounded by the selection window, so latency should be comparatively flat in density and low in variance. NR-V2X Mode 2 should resemble DSRC in access latency while improving reliability through retransmission, and the Hybrid path should trade higher latency for near-complete coverage, because the network route fills the NLOS gaps that the direct link cannot bridge.

Two caveats bound the interpretation. First, because the physical layer is reduced to a SINR-to-error mapping, the absolute delivery ratios carry more uncertainty than the latency ordering, which follows almost entirely from the access mechanism and is therefore more trustworthy. Second, the safety metrics depend on the surrogate conflict definition used by the oracle; they are best read comparatively how much sooner one technology delivers a warning than another rather than as absolute collision counts. With those bounds in mind, the exercise served its purpose: it turned a set of protocol descriptions into a working mechanism whose behaviour could be interrogated, and it made concrete the single point that most affects intersection safety that the distribution's tail, and the run-length of losses within it, matter more than the average.

DSRC vs C-V2X - intersection collision-avoidance comparison

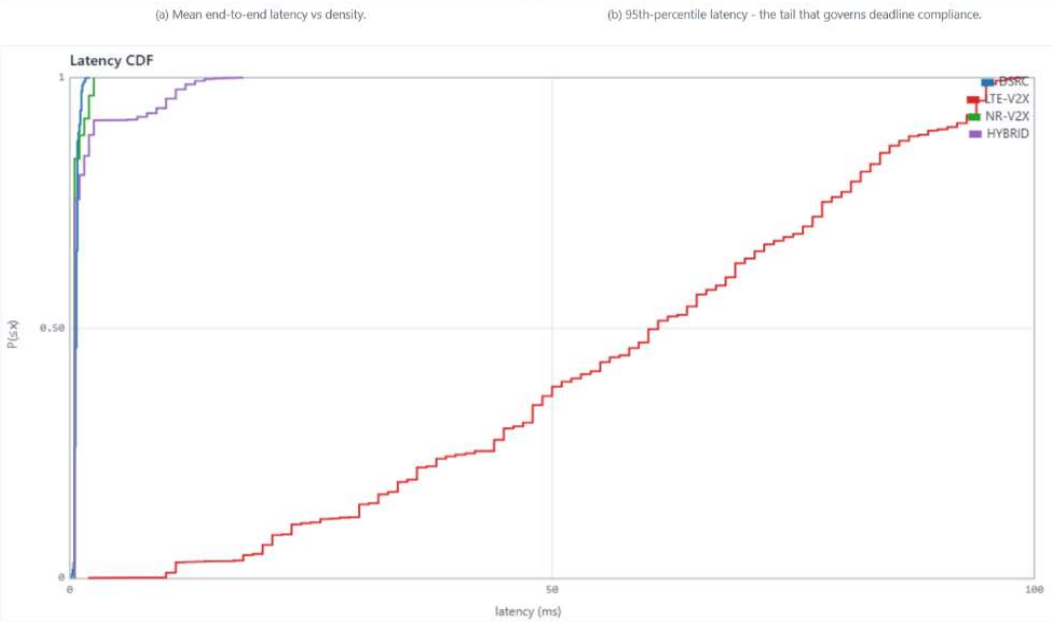
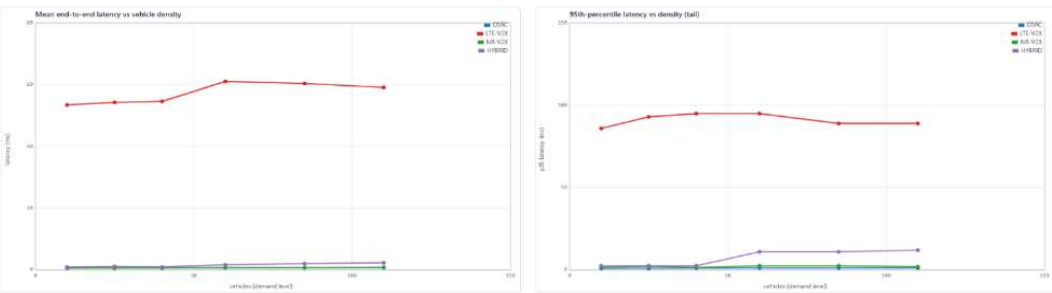
Generated 2026-07-20 00:48 UTC · seed 42 · environment: clear, RSU on, background load 0.0% · each point = 14 s headless run · densities 10, 25, 40, 60, 85, 110 vehicles.

Self-contained report produced by the V2X intersection simulator. Every RAT was run through the identical traffic, geometry and channel model; only the access-layer/scheduler differs. Figures are indicative of relative behaviour - see the tool README for model assumptions and simplifications.

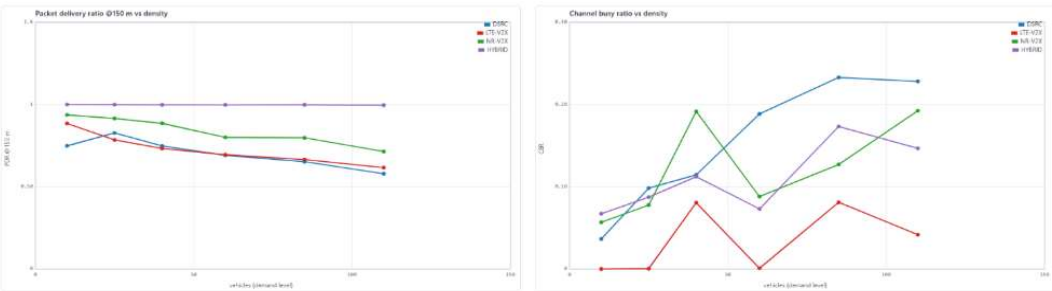
Summary at 60 vehicles

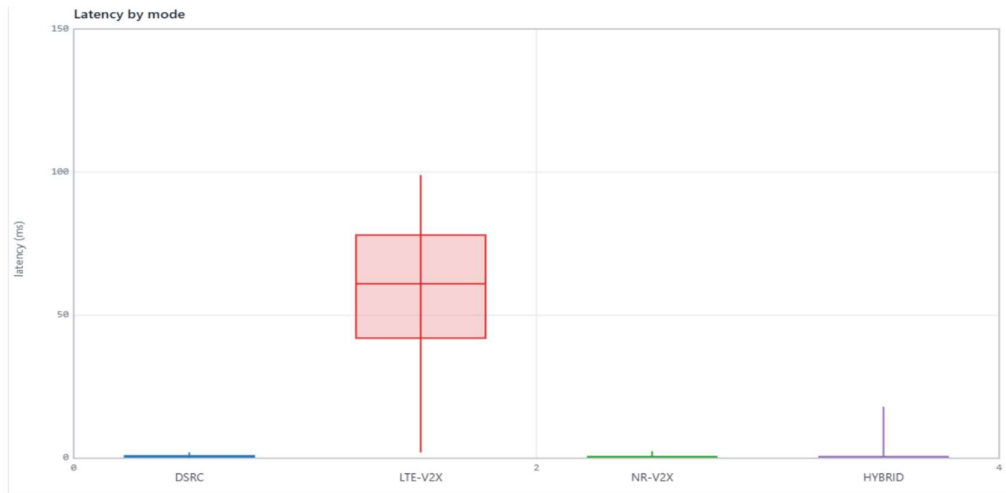
RAT	PDR@150 m	Latency (ms)	p95 (ms)	Jitter (ms)	CBR	Goodput (kbps)	ICW on-time	Warn delay (ms)
DSRC	69.1%	0.71	1.20	0.00	18.9%	28821	427	0.58
LTE-V2X	69.5%	61.0	95.0	0.68	0.1%	19067	426	22.9
NR-V2X	80.0%	0.74	2.50	0.00	8.8%	30328	384	1.16
HYBRID	99.8%	1.66	11.0	0.00	7.3%	34122	420	2.40

Latency



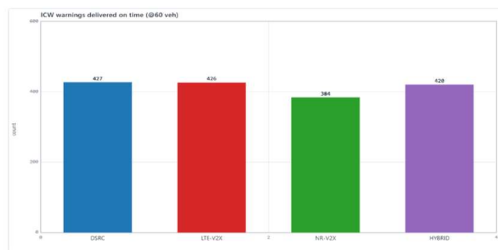
Reliability & load



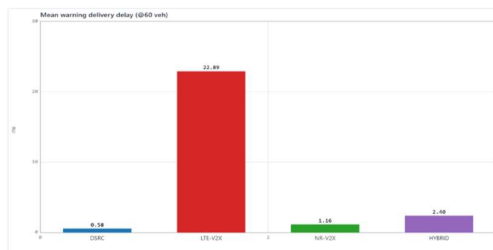


(f) Per-RAT latency distribution (box: quartiles; whiskers: min/max).

Safety-relevant delivery



(g) Intersection Collision Warnings delivered before the braking deadline.



(h) Mean warning delivery delay by RAT.

Interpretation

Expected contrast (verify against the figures above): DSRC and NR-V2X provide millisecond access latency; LTE-V2X Mode 4 sits in the tens-of-milliseconds range with a heavier upper tail from persistent SPS resource collisions; PDR falls with NLOS distance and rising CBR for all RATs; the Hybrid PC5+Uu path trades higher latency for near-complete coverage because the network fills NLOS gaps. Because it is the latency *tail* (panel c) and the run-length of losses, not the mean, that decides whether a safety deadline is met, panels (b), (c) and (g) are the operative ones for collision avoidance.

Full sweep data

Mode	Veh	PDR@150	Lat	p95	p99	Jitter	CBR	Goodput
DSRC	10	74.9%	0.60	0.90	1.00	0.04	3.7%	1069
DSRC	25	82.7%	0.64	0.90	1.30	0.00	9.8%	6149
DSRC	40	74.8%	0.68	1.12	1.30	0.01	11.4%	13861
DSRC	60	69.1%	0.71	1.20	1.40	0.00	18.9%	28821
DSRC	85	65.2%	0.73	1.20	1.40	0.00	23.3%	44476
DSRC	110	58.0%	0.82	1.20	2.20	0.00	22.8%	63569
LTE-V2X	10	88.5%	53.5	86.0	91.0	9.50	0.0%	1063
LTE-V2X	25	78.5%	54.3	93.0	95.0	4.86	0.0%	5408
LTE-V2X	40	73.3%	54.6	95.0	98.0	3.17	8.1%	10764
LTE-V2X	60	69.5%	61.0	95.0	96.0	0.68	0.1%	19067
LTE-V2X	85	66.5%	60.4	89.0	96.0	0.58	8.1%	23896
LTE-V2X	110	61.6%	59.1	89.0	95.0	2.87	4.2%	36925
NR-V2X	10	93.7%	0.61	1.50	2.50	0.01	5.7%	1312
NR-V2X	25	91.5%	0.63	2.00	2.50	0.02	7.8%	6175
NR-V2X	40	88.6%	0.60	1.50	2.50	0.05	19.1%	14500
NR-V2X	60	80.0%	0.74	2.50	2.50	0.00	8.8%	30328
NR-V2X	85	79.8%	0.75	2.50	2.50	0.00	12.7%	51018
NR-V2X	110	71.4%	0.75	2.00	2.50	0.00	19.2%	76498
HYBRID	10	100.0%	0.95	2.50	11.0	0.90	6.7%	1309
HYBRID	25	99.9%	1.11	2.50	12.0	0.07	8.7%	6766
HYBRID	40	99.8%	0.98	2.50	12.0	0.21	11.2%	14960
HYBRID	60	99.8%	1.66	11.0	13.0	0.00	7.3%	34122
HYBRID	85	99.8%	2.06	11.0	14.0	0.06	17.3%	55648
HYBRID	110	99.6%	2.29	12.0	13.0	0.69	14.7%	89280

[Download raw CSV](#)

Parameters: mode-independent settings held constant - beacon 10 Hz, packet 300 B, awareness 150 m, Tx 23 dBm, DCC/CR on, RRI 100 ms, blind-retx 1.

12. V2X CYBERSECURITY: THREATS AND COUNTERMEASURES

12.1 Introduction

This section summarises the review carried out during the internship and written up separately as *“From Spoofing to Denial-of-Service: A Curriculum-Integrated Taxonomy of V2X Cybersecurity Threats and Countermeasures”*.

The connection to the preceding material is not incidental. Section 5 argued that cooperative perception requires an agent to act on information it did not itself observe. That is a statement about trust, and it means every channel opened for cooperation is also a channel opened for attack: a vehicle that fuses a neighbour’s feature map into its own scene representation has, in effect, granted that neighbour write access to its perception of the world.

The taxonomy is organised around the classical CIA triad, extended for the vehicular context with authenticity, availability and privacy. Safety-critical vehicular communication inverts the usual priority order: availability and integrity matter far more than confidentiality, because a safety message is broadcast to everyone by design, and the damage from a forged or suppressed message is immediate and physical.

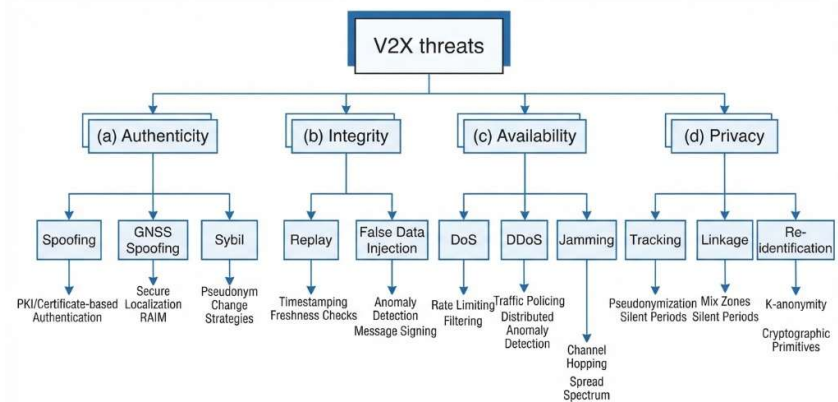


Fig. 12.1 Taxonomy of V2X cybersecurity threats.

12.2 Attacks on Authenticity

Spoofing. The attacker transmits messages purporting to come from a vehicle other than itself, or from a vehicle that does not exist. The classic instance is a phantom vehicle: an attacker broadcasts messages describing a vehicle braking hard directly ahead of a victim, and the victim’s collision-avoidance system responds. Petit and Shladover’s analysis of potential cyberattacks on automated vehicles [38] identifies this class as among the most consequential, precisely because the victim’s response is automatic.

GNSS spoofing is the variant that connects most directly to the rest of this report. By transmitting counterfeit satellite signals, an attacker can cause a receiver to compute a false position and, since a GNSS receiver derives time from the same signals, a false position implies a false clock. The consequences run straight through Section 3: the victim's timestamps become wrong, its estimate of Δt for every message becomes wrong, and any temporal compensation is applied by the wrong amount, silently. This is the sharpest illustration of why FreeAlign's approach [34] recovering the spatio-temporal relationship between agents from the geometry of what they see rather than from an external signal is interesting for reasons that go beyond convenience.

Sybil attacks [44] fabricate multiple identities to simulate the presence of many vehicles. The attack is powerful because so many V2X mechanisms are consensus-based: a claim corroborated by ten vehicles is more credible than a claim made by one, and an attacker who can be those ten vehicles subverts the corroboration. It is particularly damaging for cooperative perception, where a fabricated majority reporting a phantom obstacle can override the ego vehicle's own correct but lone observation that the road is clear.

12.3 Countermeasure: PKI and IEEE 1609.2

The standardised response to attacks on authenticity is a Public Key Infrastructure, specified for the WAVE stack in IEEE 1609.2 [3] and, in the European profile, by ETSI TS 102 941 [45]. Every message is digitally signed with a private key whose certificate is issued by a trusted authority, and a receiver verifies the signature and certificate chain before accepting the message. Raya and Hubaux's foundational treatment [39] established the design pattern the standards subsequently adopted.

Three properties are worth noting. **Pseudonym certificates**: a vehicle holds a pool of short-lived certificates and rotates between them, so its messages cannot be trivially linked into a trajectory the mechanism by which the standard reconciles authentication with privacy (Section 12.5). **Certificate revocation**: a misbehaving vehicle can have its certificates revoked, though distributing revocation information promptly to a national fleet is a hard and only partially solved problem. **Verification cost**: every verification costs computation, and a receiver in dense traffic must perform hundreds per second which is not merely a performance issue but an attack surface (Section 12.4).

The critical limitation, and the one that motivates Section 12.6, is that **PKI authenticates the sender**, not the content. It establishes that a message genuinely came from a certified vehicle, and says nothing about whether that vehicle is telling the truth. A legitimately certified vehicle whose software has been compromised passes every cryptographic check while transmitting entirely fabricated data.

12.4 Attacks on Integrity and Availability

Replay. The attacker captures a legitimate, correctly signed message and retransmits it later. Every cryptographic check passes, because the message is genuine it is merely no longer true. A recorded emergency-brake warning replayed into free-flowing traffic is a valid message describing a condition that no longer exists. The countermeasure is freshness verification against the message timestamp and note what this implies: freshness checking is entirely dependent on clock synchronisation. A vehicle whose clock has drifted, or whose GNSS time has been spoofed, will compute the wrong age for every message it receives, and may accept a replayed message as fresh. The clock is a security asset, not merely an engineering convenience.

False data injection is the insider attack: a legitimately certified vehicle transmits false content a position it is not at, or a detected object that does not exist. Cryptography cannot address this, and its severity grows with the level of collaboration (Section 5.3). Injecting a false object into a CPM is bad; injecting a maliciously crafted feature map into an intermediate-fusion pipeline is worse, because the receiver has no interpretable representation to sanity-check it cannot look at a tensor and see that it is implausible in the way it can look at a claimed object and see that its own LiDAR shows empty road there.

Denial of service. The vehicular network is unusually exposed, for a reason that follows directly from Section 3.8: the channel is shared and saturates well before its nominal capacity, so an attacker who floods it need not be sophisticated, merely loud. A more economical variant exploits the verification cost of Section 12.3: generating a message with an invalid signature is computationally trivial for the attacker, while verifying that it is invalid costs the victim a full elliptic-curve verification. A receiver forced to spend its entire processing budget rejecting forged messages has none left for genuine ones a denial of service achieved through the security mechanism itself. Countermeasures are partial: rate limiting, plausibility pre-filtering before the expensive cryptographic check, and hardware acceleration.

Jamming is denial of service at the radio layer: the attacker raises the noise floor until the signal-to-noise ratio of legitimate transmissions falls below the receiver's threshold. It is straightforward to mount with inexpensive software-defined radio hardware and difficult to prevent. Its relevance here is that it produces exactly the interruption pattern of Section 3.7 a sustained, correlated loss of all packets from all senders so a system with robust interruption handling [31] degrades gracefully under jamming while one without it does not. At this point robustness and security are the same property.

12.5 Privacy

A vehicle broadcasting its position ten times a second is, from a privacy standpoint, a beacon. If those broadcasts can be linked, an observer with a modest number of receivers can reconstruct an individual's movements. The **pseudonym** mechanism exists to break this linkage but is not a complete solution: if a vehicle switches pseudonym while its position,

speed and heading are being continuously observed, an adversary trivially associates the old identity with the new. The mitigations changing pseudonyms inside a mix zone where many vehicles are close together, or during a silent period in which the vehicle transmits nothing both trade safety for privacy, since a silent vehicle is an invisible one. This is a genuine, irreducible tension rather than a design oversight: the same property that makes a safety message useful that it truthfully and continuously reveals where a vehicle is what makes it a privacy hazard.

12.6 Future Security Challenges

Adversarial attacks on the shared representation. As cooperative perception moves from object-level to feature-level sharing, the attack surface moves with it. An adversarial perturbation crafted against a neural encoder need not look implausible to a physics-based plausibility filter, because it is not a claim about the world but a tensor engineered to make a downstream network produce a particular wrong answer.

Scalability of the credential system. A national fleet of tens of millions of vehicles, each holding a rotating pool of short-lived certificates, generates certificate-management and revocation-distribution problems of a scale not yet operationally demonstrated.

Post-quantum cryptography. The signature scheme in IEEE 1609.2 is elliptic-curve based, and elliptic-curve signatures are broken by a sufficiently large quantum computer. Vehicles have service lives of fifteen to twenty years, so vehicles certified today may still be on the road when the threat becomes real. Post-quantum signature schemes have substantially larger signatures and keys, which collides directly with the bandwidth constraint of Section 3.8.

Security of the compensation layer itself. A prediction module is a target. An attacker who understands how a receiver extrapolates features forward could craft a sequence of messages designed to make the prediction diverge an attack on the dynamics of the perception system rather than on any single message.

13. INTERNSHIP OUTCOMES

13.1 Knowledge Gained

V2X communication. Starting from no prior exposure, a working understanding was developed of the four communication modes, the standardised message sets, and the two radio access families. The most useful realisation was structural rather than factual: DSRC and C-V2X are not competing implementations of one idea but embodiments of two different philosophies of medium access decentralised contention versus reservation and almost every difference in their observed behaviour follows from that one choice.

Communication impairments. The decomposition of end-to-end latency into its components was the most valuable analytical tool acquired: it makes immediately clear which terms matter (queueing, and for feature-level sharing transmission) and which do not (propagation, by four orders of magnitude), and therefore where to look for improvement.

Synchronisation. The recognition that the total temporal misalignment between two agents is the sum of the communication delay, the clock offset and the sensor sampling-phase offset and that only the first is what is usually meant by “latency” was the single insight that most changed the author’s understanding of the field.

Cooperative perception. An understanding of the levels of collaboration and of why the field converged on intermediate fusion; of the pipeline from perception through communication, compensation and fusion to decision; and of why the compensation layer, which is absent from most introductory treatments, is not optional.

Simulation frameworks and datasets. A clear picture of what each tool models, what it does not, and of the gap between the perception tools, which take delay as a parameter, and the network tools, which produce it as a result.

Cybersecurity. A structured view of the V2X threat landscape and the more interesting outcome of the ways in which it is entangled with the perception problem: the clock is a security asset; a confidence estimator and a misbehaviour detector are structurally the same computation.

Research methodology. Two lessons stand out. That the assumptions of a paper are more informative than its results, so a table comparing assumptions is worth more than a table comparing accuracies. And that evaluating on a metric which is a proxy for the thing one cares about offline average precision rather than driving behaviour can conceal failure, as [35] demonstrates.

13.2 Technical Skills

Reading and interpreting standards documents (IEEE 802.11p, IEEE 1609.2, SAE J2735, 3GPP specifications); reading a research paper for its assumptions rather than only for its method

and results; constructing a simple discrete-event simulation in Python, and discovering how much a protocol description leaves unstated until one tries to implement it; data analysis and plotting; and technical writing in a formal register, including the construction of comparison tables that expose structure rather than merely list facts.

Table 13.1 Mapping of internship objectives to outcomes.

Objective	Where addressed	Outcome
O1 - Understand V2X communication	Section 2	Working knowledge of modes, message sets and both radio families
O2 - Characterise communication impairments	Section 3	Delay decomposition; quantified clock drift; loss taxonomy; bandwidth arithmetic
O3 - Understand their effect on perception	Section 4	Spatial/semantic misalignment distinction; displacement-error analysis
O4 - Survey ML methods for prediction	Section 6	Predictive modeling insights
O5 -Study of research papers	Throughout the report (Sections 2-12)	Literature review and synthesis
O6 - Familiarise with frameworks and datasets	Sections 9, 10	Capability/limitation map of six frameworks and three datasets
O7 - Compare DSRC and C-V2X	Section 11	Educational simulation exercise; access mechanisms understood in detail
O8 - Review V2X cybersecurity	Section 12	Taxonomy of attacks and countermeasures; link to compensation established

13.3 What the Internship Did Not Produce

Stated explicitly, in keeping with the declaration at the front of this report: no novel algorithm was designed, no model was trained, and no result reported here should be understood as a research contribution. The simulation exercise of Section 11 is a learning aid. The value of the internship lies in the understanding acquired and documented here, and in the identification in Section 15 of where the open problems are.

14. FUTURE WORK

The directions below follow from gaps identified during the study. They are offered as the questions the author would pursue if the work were continued, not as claimed contributions.

Delay compensation under an uncertain delay estimate. Almost every method in Section 6 and 7 takes Δt as a known quantity, while Section 3.5 shows it is not: it is contaminated by clock drift, and the contamination is silent. A compensation module that accepted a distribution over Δt rather than a point estimate, and marginalised over it, would be a direct extension of CoDynTrust’s uncertainty framework [33].

Adaptive synchronisation. FreeAlign [34] recovers pose and time from object geometry but needs co-viewed objects; GNSS provides both but fails in tunnels and can be spoofed. A system that used GNSS when it was trustworthy and fell back to geometric self-alignment when it was not and that could detect which regime it was in would be more robust than either alone. Relatedly, a vehicle knows things about its own clock (when it last synchronised, how long its GNSS has been in outage), and transmitting a confidence in its own timestamp alongside the timestamp would let receivers weight its contributions accordingly, at a cost of a few bytes.

Co-simulation rather than parameter injection. The gap identified in Section 9.7 that perception tools take delay as a parameter while network tools produce it as a result is a concrete and tractable engineering project. Coupling OpenCOOD to a Veins or ns-3 channel model, so that the delay a compensation module must handle is the delay the protocol stack actually produced under the density SUMO actually generated, would allow the compensation literature to be evaluated under realistic conditions.

6G-enabled perception. If the network can be made task-aware (Section 8.2), the feature-selection problem can be solved jointly with resource allocation rather than separately from it.

Foundation models. Large pre-trained models offer, in principle, a stronger prior on how traffic scenes evolve than any model trained on a cooperative perception dataset alone which is what a compensation module needs, since compensation is the application of a prior about the future. Two obstacles are immediate: inference latency, since a model that takes 200 ms to run has consumed the budget it was meant to protect; and the risk of confident hallucination, which Section 6.5 argues is the worst possible failure mode here.

Edge AI and predictive communication. An edge server fusing observations from many vehicles avoids the pairwise-exchange cost of the fully distributed case, at the price of a round trip and a single point of failure; the trade-off is quantifiable and appears not to have been quantified carefully. Separately, if the receiver can predict what the sender will send, the sender need only transmit the residual a well-established idea in video coding that appears not yet to have been applied to cooperative feature sharing, though FFNet’s feature-flow transmission [30] is a step in this direction.

15. CONCLUSION

This internship set out to build a technically grounded understanding of Vehicle-to-Everything communication and of the cooperative perception systems built on top of it.

V2X is best understood as a response to a geometric fact: a vehicle's sensors see only what is in their line of sight, and much of what matters in traffic is not. The standards that have grown up around this fact DSRC and IEEE 802.11p on one side, LTE-V2X and NR-V2X on the other differ less in their goals than in their answer to a single question, namely how a shared and finite medium should be arbitrated among participants who cannot coordinate. Almost every difference in their behaviour follows from that choice, and the evolution now under way towards 6G promises to change the terms of the trade-off without abolishing it.

The link that these standards provide is, in practice, an imperfect one. Messages are delayed by tens to hundreds of milliseconds, dominated not by the speed of light but by contention for the medium and for the large payloads that feature-level cooperation requires by the time taken to serialise them. The delay varies unpredictably. Packets are lost, sometimes in sustained bursts. Clocks drift silently, so that even the age of a message is uncertain. Sensors on different vehicles are triggered at different instants, so misalignment would exist even with a perfect network. And the channel is shared and finite, saturating precisely when traffic is densest and the safety application is needed most.

Carried into the perception pipeline, these properties mean that a delayed observation describes a world that has moved on. The empirical evidence reviewed here goes further, and is the most important thing the author learned: uncompensated cooperative perception can be worse than no cooperative perception at all, because a fusion module that is not told its input is stale will assert with confidence that an object is where it is not. Compensation is therefore not a refinement of the architecture but a required component of it, and the literature is best understood as a set of different answers to the same question warping features along an estimated motion field, transmitting a temporal gradient so the receiver can extrapolate, reconstructing current-time features by attending backwards along predicted trajectories, recovering the timing relationship from the geometry of what the agents see, or - most valuably, in the author's judgement concentrating less on predicting accurately and more on knowing how much to trust the prediction.

No new algorithm was produced during this internship, and none is claimed. What was produced is an understanding of a field of where its difficulty actually resides, of which of its apparent difficulties are illusory and which are underestimated, and of the gap between the communications literature, which knows what the channel does, and the perception literature, which mostly assumes. That gap seems to the author to be where the interesting work lies.

REFERENCES

- [1] J. B. Kenney, "Dedicated Short-Range Communications (DSRC) Standards in the United States," in *Proceedings of the IEEE*, vol. 99, no. 7, pp. 1162-1182, July 2011.
- [2] *IEEE Standard for Information Technology - Local and Metropolitan Area Networks; Part 11: Wireless LAN MAC and PHY Specifications; Amendment 6: Wireless Access in Vehicular Environments*, IEEE Std 802.11p-2010, 2010.
- [3] *IEEE Standard for Wireless Access in Vehicular Environments - Security Services for Applications and Management Messages*, IEEE Std 1609.2-2016, 2016.
- [4] *IEEE Standard for Wireless Access in Vehicular Environments (WAVE) - Multi-Channel Operation*, IEEE Std 1609.4-2016, 2016.
- [5] *Dedicated Short Range Communications (DSRC) Message Set Dictionary*, SAE Standard J2735, SAE International.
- [6] *On-Board System Requirements for V2V Safety Communications*, SAE Standard J2945/1, SAE International.
- [7] *Intelligent Transport Systems (ITS); Vehicular Communications; Basic Set of Applications; Part 2: Cooperative Awareness Basic Service*, ETSI EN 302 637-2.
- [8] *Intelligent Transport Systems (ITS); Vehicular Communications; Basic Set of Applications; Part 3: Decentralized Environmental Notification Basic Service*, ETSI EN 302 637-3.
- [9] *Intelligent Transport Systems (ITS); Vehicular Communications; Collective Perception Service*, ETSI TS 103 324.
- [10] *Service Requirements for V2X Services*, 3GPP TS 22.185, Release 14.
- [11] R. Molina-Masegosa and J. Gozalvez, "LTE-V for Sidelink 5G V2X Vehicular Communications: A New 5G Technology for Short-Range Vehicle-to-Everything Communications," in *IEEE Vehicular Technology Magazine*, vol. 12, no. 4, pp. 30-39, Dec. 2017.
- [12] *Architecture Enhancements for 5G System (5GS) to Support Vehicle-to-Everything (V2X) Services*, 3GPP TS 23.287, Release 16.
- [13] G. Naik, B. Choudhury and J. -M. Park, "IEEE 802.11bd & 5G NR V2X: Evolution of Radio Access Technologies for V2X Communications," in *IEEE Access*, vol. 7, pp. 70169-70184, 2019.
- [14] M. H. C. Garcia *et al.*, "A Tutorial on 5G NR V2X Communications," in *IEEE Communications Surveys & Tutorials*, vol. 23, no. 3, pp. 1972-2026, thirdquarter 2021.
- [15] W. Saad, M. Bennis and M. Chen, "A Vision of 6G Wireless Systems: Applications, Trends, Technologies, and Open Research Problems," in *IEEE Network*, vol. 34, no. 3, pp. 134-142, May/June 2020.
- [16] M. Noor-A-Rahim *et al.*, "6G for Vehicle-to-Everything (V2X) Communications: Enabling Technologies, Challenges, and Opportunities," in *Proceedings of the IEEE*, vol. 110, no. 6, pp. 712-734, June 2022.
- [17] D. L. Mills, "Internet time synchronization: the network time protocol," in *IEEE Transactions on Communications*, vol. 39, no. 10, pp. 1482-1493, Oct. 1991.
- [18] *IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems*, IEEE Std 1588-2019, 2019.

- [19] O. Kaiwartya *et al.*, "Internet of Vehicles: Motivation, Layered Architecture, Network Model, Challenges, and Future Aspects," in *IEEE Access*, vol. 4, pp. 5356-5373, 2016.
- [20] Y. Han, H. Zhang, H. Li, Y. Jin, C. Lang and Y. Li, "Collaborative Perception in Autonomous Driving: Methods, Datasets, and Challenges," in *IEEE Intelligent Transportation Systems Magazine*, vol. 15, no. 6, pp. 131-151, Nov.-Dec. 2023.
- [21] Wang, TH., Manivasagam, S., Liang, M., Yang, B., Zeng, W., Urtasun, R. (2020). V2VNet: Vehicle-to-Vehicle Communication for Joint Perception and Prediction. In: Vedaldi, A., Bischof, H., Brox, T., Frahm, JM. (eds) Computer Vision – ECCV 2020. ECCV 2020.
- [22] R. Xu, H. Xiang, X. Xia, X. Han, J. Li and J. Ma, "OPV2V: An Open Benchmark Dataset and Fusion Pipeline for Perception with Vehicle-to-Vehicle Communication," *2022 International Conference on Robotics and Automation (ICRA)*, Philadelphia, PA, USA, 2022.
- [23] Xu, R., Xiang, H., Tu, Z., Xia, X., Yang, MH., Ma, J. (2022). V2X-ViT: Vehicle-to-Everything Cooperative Perception with Vision Transformer. In: Avidan, S., Brostow, G., Cissé, M., Farinella, G.M., Hassner, T. (eds) Computer Vision – ECCV 2022. ECCV 2022.
- [24] H. Yu, Y. Luo, M. Shu, Y. Huo, Z. Yang, Y. Shi, Z. Guo, H. Li, X. Hu, J. Yuan, and Z. Nie, "DAIR-V2X: A large-scale dataset for vehicle-infrastructure cooperative 3D object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 21361–21370.
- [25] H. Yu, W. Yang, H. Ruan, Z. Yang, Y. Tang, X. Gao, X. Hao, Y. Shi, Y. Pan, N. Sun, J. Song, J. Yuan, P. Luo, and Z. Nie, "V2X-Seq: A large-scale sequential dataset for vehicle-infrastructure cooperative perception and forecasting," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 5486–5495.
- [26] Y. Hu, S. Fang, Z. Lei, Y. Zhong, and S. Chen, "Where2comm: Communication-efficient collaborative perception via spatial confidence maps," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 35, 2022, pp. 4874–4886.
- [27] Y. Lu, Q. Li, B. Liu, M. Dianati, C. Feng, S. Chen, and Y. Wang, "Robust collaborative 3D object detection in presence of pose errors," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2023, pp. 4812–4818.
- [28] Lei, Z., Ren, S., Hu, Y., Zhang, W., Chen, S. (2022). Latency-Aware Collaborative Perception. In: Avidan, S., Brostow, G., Cissé, M., Farinella, G.M., Hassner, T. (eds) Computer Vision – ECCV 2022. ECCV 2022.
- [29] S. Wei, Y. Wei, Y. Hu, Y. Lu, Y. Zhong, S. Chen, and Y. Zhang, "Asynchrony-robust collaborative perception via bird's eye view flow," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 36, Dec. 2023, pp. 28462–28477.
- [30] H. Yu, Y. Tang, E. Xie, J. Mao, P. Luo, and Z. Nie, "Flow-based feature fusion for vehicle-infrastructure cooperative 3D object detection," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 36, Dec. 2023.
- [31] S. Ren *et al.*, "Interruption-Aware Cooperative Perception for V2X Communication-Aided Autonomous Driving," in *IEEE Transactions on Intelligent Vehicles*, vol. 9, no. 4, pp. 4698-4714, April 2024.
- [32] Z. Song, L. Yang, F. Wen, and J. Li, "TraF-Align: Trajectory-aware feature alignment for asynchronous multi-agent perception," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 12048–12057.
- [33] Y. Xu *et al.*, "CoDynTrust: Robust Asynchronous Collaborative Perception via Dynamic Feature Trust Modulus," *2025 IEEE International Conference on Robotics and Automation (ICRA)*, Atlanta, GA, USA, 2025, pp. 336-342.

- [34] Z. Lei *et al.*, "Robust Collaborative Perception without External Localization and Clock Devices," 2024 *IEEE International Conference on Robotics and Automation (ICRA)*, Yokohama, Japan, 2024.
- [35] A. Guo *et al.*, "When autonomous vehicle meets V2X cooperative perception: How far are we?," in *Proc. 40th IEEE/ACM Int. Conf. Automated Software Engineering (ASE)*, 2025. (arXiv:2509.24927.)
- [36] J. Wang and T. Nordström, "Latency robust cooperative perception using asynchronous feature fusion," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Feb. 2025, pp. 4862–4871.
- [37] W. Wang, J. Wang, Y. Deng, and G. Tan, "BEVSync: Asynchronous data alignment for camera-based vehicle-infrastructure cooperative perception under uncertain delays," in *Proc. AAAI Conf. Artif. Intell.*, vol. 39, no. 14, 2025, pp. 14699–14707.
- [38] J. Petit and S. E. Shladover, "Potential cyberattacks on automated vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 16, no. 2, pp. 546–556, Apr. 2015.
- [39] M. Raya and J.-P. Hubaux, "Securing vehicular ad hoc networks," *J. Comput. Secur.*, vol. 15, no. 1, pp. 39–68, 2007.
- [44] J. R. Douceur, "The Sybil attack," in *Proc. 1st Int. Workshop on Peer-to-Peer Systems (IPTPS)*, 2002, pp. 251–260.
- [45] *Intelligent Transport Systems (ITS); Security; Trust and Privacy Management*, ETSI TS 102 941.
- [46] C. Sommer, R. German, and F. Dressler, "Bidirectionally coupled network and road traffic simulation for improved IVC analysis," *IEEE Trans. Mobile Comput.*, vol. 10, no. 1, pp. 3–15, Jan. 2011.
- [47] A. Varga and R. Hornig, "An overview of the OMNeT++ simulation environment," in *Proc. 1st Int. Conf. Simulation Tools and Techniques (SIMUTools)*, 2008.
- [48] P. A. Lopez, M. Behrisch, L. Bieker-Walz, J. Erdmann, Y.-P. Flötteröd, R. Hilbrich, L. Lücken, J. Rummel, P. Wagner, and E. Wiessner, "Microscopic traffic simulation using SUMO," in *Proc. 21st IEEE Int. Conf. Intell. Transp. Syst. (ITSC)*, 2018, pp. 2575–2582.
- [49] A. Dosovitskiy, G. Ros, F. Codevilla, A. López, and V. Koltun, "CARLA: An open urban driving simulator," in *Proc. 1st Conf. Robot Learning (CoRL)*, 2017, pp. 1–16.
- [50] T. R. Henderson, M. Lacage, and G. F. Riley, "Network simulations with the ns-3 simulator," in *Proc. ACM SIGCOMM (Demonstration)*, 2008.
- [51] A. H. Lang, S. Vora, H. Caesar, L. Zhou, J. Yang, and O. Beijbom, "PointPillars: Fast encoders for object detection from point clouds," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 12697–12705.
- [52] Y. Zhou and O. Tuzel, "VoxelNet: End-to-end learning for point cloud based 3D object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2018, pp. 4490–4499.
- [53] A. Bazzi, B. M. Masini, A. Zanella, and I. Thibault, "On the performance of IEEE 802.11p and LTE-V2V for the cooperative awareness of connected vehicles," *IEEE Trans. Veh. Technol.*, vol. 66, no. 11, pp. 10419–10432, Nov. 2017.
- [54] A. Bazzi, G. Cecchini, A. Zanella, and B. M. Masini, "Study of the impact of PHY and MAC parameters in 3GPP C-V2V mode 4," *IEEE Access*, vol. 6, pp. 71685–71698, 2018.
- [55] 5G Automotive Association (5GAA), "V2X functional and performance test report: Test procedures and results," 5GAA White Paper, 2018.

Week	Activities	Output
1	Orientation; introduction to ITS and V2X; foundational surveys	Reading notes; Section 1 outline
2	V2X standards: IEEE 802.11p, IEEE 1609 family, SAE J2735	Section 2 draft
	Detailed study of the twenty cooperative perception papers	Throughout the report (Sections 2-12)
3	C-V2X: LTE-V2X sidelink, SPS, NR-V2X	Section 2 completed
4	Communication impairments: delay, loss, clock drift, bandwidth	Section 3 draft
	Cooperative perception: fusion levels, pipeline	Sections 4–5 draft
5	Machine learning for temporal prediction	Sections 6–7 draft
6	Simulation frameworks: SUMO, OMNeT++, Veins, ns-3, CARLA, OpenCOOD	Section 9 draft
	Public datasets: OPV2V, V2XSet, DAIR-V2X	Section 10 draft
7	DSRC / C-V2X simulation exercise: implementation and runs	Section 11
	Cybersecurity review: taxonomy of threats and countermeasures	Review paper draft; Section 12
8	Consolidation; report writing; presentation to the guide	Final report